

21-701: Discrete Mathematics

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These notes are based on the CMU graduate class 21-701: Discrete Mathematics that I took in Fall 2024 with Prof. Florian Frick. The primary value to be gained from the content of these notes is the techniques used in proving the results, rather than the statements of the results themselves.

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1 Probabilistic Combinatorics

1.1 A Motivating Example

The first question we will consider is about coloring hypergraphs.

Definition 1.1. For an integer $k \geq 2$, a k -uniform hypergraph is a set H of k -element sets.

The elements of H represent the edges of the hypergraph. The vertices (or ground set) are given by $V = \bigcup_{e \in H} e$. We may choose to specify that a hypergraph has a ground set V , in which case we only require that $\bigcup_{e \in H} e \subseteq V$ (so that $H \subseteq \binom{V}{k}$). Although these two definitions are not the same, our problems where we specify a ground set will often involve maximizing or minimizing a quantity on a given number of vertices, and so taking $\bigcup_{e \in H} e \subsetneq V$ will be suboptimal. Because of this, we will choose not to take much issue with these definitions.

Definition 1.2. Let H be a k -uniform hypergraph on the ground set V . For $c \geq 1$, a c -coloring of H is a function $f : V \rightarrow [c]$ such that every edge receives at least two colors (i.e. $|f(e)| > 1$ for all $e \in H$).

We will write $m(k)$ for the smallest number of edges of a k -uniform hypergraph H (on any vertex set) that does not admit a 2-coloring. For example, $m(2) = 3$: any (simple) graph with ≤ 2 edges can be 2-colored, but K_3 (which has 3 edges) cannot. The problem of computing $m(3)$ is much harder.

Proposition 1.3. $m(3) \geq 7$.

Proof. Let H be a 3-uniform hypergraph with $|H| \leq 6$. We want to show that H is 2-colorable. We'll first show that we only have to consider the case when the vertex set V has at most 6 elements.

Suppose that $|V| \geq 7$. We'll show that there are distinct $v, w \in V$ that do not have any common edges between them. There are $\leq 6 \cdot \binom{3}{2} = 18$ pairs of vertices in V that can be connected by an edge in H , and $\geq \binom{7}{2} = 21$ pairs of vertices in V . Thus, there must be some pair of vertices v and w that is not connected by any edges in H . We may merge v and w (formally, define a new hypergraph $H' = \{e \in H : w \notin e\} \cup \{(e \setminus \{w\}) \cup \{v\} : e \in H, w \in e\}$). Any 2-coloring on the merged graph induces a 2-coloring on H . Repeating this process, we only need to consider the case when $|V| \leq 6$.

Now, the worst case we can possibly have is $|H| = 6$ and $|V| = 6$ (if there are < 6 edges, add some random edges, and if there are < 6 vertices, add some vertices not connected to anything). We'll show that a 2-coloring must exist. We will consider the number of 2-colorings that have exactly 3 vertices of each color. There are $\binom{6}{3} = 20$ such 2-colorings. Each edge rules out 2 of these 2-colorings, so there are $\leq 6 \cdot 2 = 12$ 2-colorings that are not allowed. This means that there must be at least one allowed 2-coloring, so that H is 2-colorable. ■

Remark 1.4. Our instinct in the above proof would be to simply count the total number of 2-colorings and the number of 2-colorings that a given edge rules out. However, there are 2^6 2-colorings and each edge rules out 2^4 2-colorings, and $6 \cdot 2^4 > 2^6$. Because of this, we had to get clever in picking which 2-colorings we counted. The reason for picking 2-colorings in which there are 3 vertices of each color is that we'd want to make sure edges are ruling out distinct colorings to minimize the double counting when we look at the total number of allowed edges.

The key idea of this proof was to show that a 2-coloring exists by counting the total number of 2-colorings, and then counting the number of 2-colorings that are not allowed, rather than to explicitly construct a 2-coloring. This non-constructive technique will be how we approach many problems.

Can we show that $m(3)$ is exactly 7? We'd want to construct a non-2-colorable graph. A similar argument to the proof above shows that we only have to consider the $|V| = 7$ case. We could modify the proof to $|H| = 6$ and $|V| = 7$ by considering colorings where there are 3 vertices colored 1 and 4 vertices colored 2: there are $\binom{7}{3} = 35$ such colorings and each edge obstructs $4 + 1 = 5$ colorings, meaning that there exists a valid coloring when $|H| \leq 6$. For this to fail in the $|H| = 7$ case, each edge would need to obstruct 5 new colorings. Noticing that disjoint edges and edges that have 2 vertices in common both rule out a common coloring, this means that every pair of edges must have exactly 1 vertex in common. The Fano Plane (Figure 1) satisfies this.

Remark 1.5. The Fano Plane is an example of a finite projective plane of order 2 (in fact, the only finite projective plane of order 2 up to labeling). In general, a finite projective plane of order $k - 1$ is a k -uniform hypergraph such that any two (distinct) vertices are contained in a unique edge and any two (distinct) edges intersect in a unique point. The vertices correspond to points and the edges to lines, so that these axioms require that every pair of points are contained in a unique line and every pair of lines intersect at a unique point. Given a finite projective plane, we can define its dual by having a vertex V_e for each edge e and an edge E_v for each vertex v (where $V_e \in E_v$ iff $v \in e$). The Fano Plane is self-dual (in that its dual can be relabelled to get the Fano Plane).

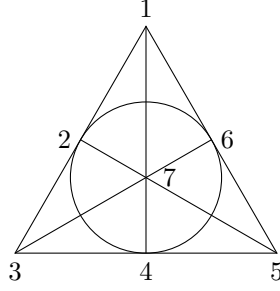


Figure 1. The Fano Plane. The 6 straight lines and the circle correspond to hyperedges of size 3. The Fano Plane is not 2-colorable, as every set of 5 vertices contains an edge, so $1 - 6$ and $2 - 5$ color splits are not possible, and each edge prohibits 5 distinct $3 - 4$ split colorings, meaning none of the $35 = 7 \cdot 5$ $3 - 4$ split colorings are possible.

We can use similar techniques to find a lower bound for $m(k)$.

Proposition 1.6. $m(k) \geq 2^{k-1}$.

Proof. Let H be a k -uniform hypergraph with $|H| < 2^{k-1}$, and let $|V| = n$. There are 2^n possible 2-colorings, and each edge obstructs $\leq 2 \cdot 2^{n-k}$ 2-colorings. Thus, there are $\leq 2^{n-k+1} \cdot |H| < 2^n$ 2-colorings that are not allowed, so there are > 0 allowed 2-colorings. In particular, there is at least 1. ■

We can rephrase this proof as follows: pick a random 2-coloring of H (by randomly coloring each vertex 1 or 2 with probability $1/2$). The, the probability that a fixed $e \in H$ is monochromatic (i.e. rules out our coloring) is $2/2^k = 1/2^{k-1}$, so the probability that *some* $e \in H$ is monochromatic is $\leq \sum_{e \in H} 1/2^{k-1} < 1$. This means that the probability that no $e \in H$ is monochromatic is > 0 , so our 2-coloring is valid with probability > 0 , meaning in particular that one must exist. This strategy of showing that a random object satisfies properties that we want with a nonzero probability, and so that some object must satisfy the properties that we want, is the key idea in probabilistic combinatorics.

1.2 A Crash Course in Probability

For us, our set Ω of possible outcomes will almost always be countable or finite. A probability function $\mathbb{P} : \Omega \rightarrow [0, 1]$ satisfies $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$. For $A \subseteq \Omega$, we define $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$.

A random variable is a function $X : \Omega \rightarrow \mathbb{R}$. The expected value of X is $\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega)$. Expectation is linear, so that $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ and $\mathbb{E}(\lambda X) = \lambda \mathbb{E}(X)$.

Proposition 1.7 (Markov's Inequality). *if X is a random variable and $a > 0$, $\mathbb{P}(X \geq a) \leq \mathbb{E}(X)/a$.*

Proof. We have that

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega) \geq \sum_{\substack{\omega \in \Omega \\ X(\omega) \geq a}} X(\omega) \mathbb{P}(\omega) \geq a \sum_{\substack{\omega \in \Omega \\ X(\omega) \geq a}} \mathbb{P}(\omega) = a \mathbb{P}(X \geq a).$$

■

The expected value of a random variable represents the center of its distribution. We can also measure how much a random variable tends to differ from its expected value: the variance of a random variable X is

$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - \mathbb{E}(X)^2.$$

More generally, the covariance of random variables X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

Variance is not linear. However, we have the following result, which is almost as good:

Proposition 1.8. *If $X = \sum_{i=1}^n X_i$ is the sum of random variables, then*

$$\text{Var}(X) = \sum_{1 \leq i, j \leq n} \text{Cov}(X_i, X_j) = \sum_{1 \leq i \leq n} \text{Var}(X_i) + \sum_{1 \leq j \leq n} \text{Cov}(X_i, X_j).$$

Proof. We have that

$$\mathbb{E}(X^2) = \mathbb{E} \left(\left(\sum_{1 \leq i \leq n} X_i \right)^2 \right) = \mathbb{E} \left(\sum_{i,j} X_i X_j \right) = \sum_{i,j} \mathbb{E}(X_i X_j),$$

and

$$\mathbb{E}(X)^2 = \mathbb{E} \left(\sum_{1 \leq i \leq n} X_i \right)^2 = \mathbb{E} \left(\sum_{1 \leq i \leq n} \mathbb{E}(X_i) \right)^2 = \sum_{1 \leq i,j \leq n} \mathbb{E}(X_i) \mathbb{E}(X_j),$$

so

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \sum_{1 \leq i,j \leq n} \mathbb{E}(X_i X_j) - \mathbb{E}(X_i) \mathbb{E}(X_j) = \sum_{1 \leq i,j \leq n} \text{Cov}(X_i, X_j).$$

The second equality follows from noting that $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$. ■

Proposition 1.9 (Chebyshev's Inequality). *If X is a random variable with finite variance and $t > 0$, then $\mathbb{P}(|X - \mathbb{E}(X)| \geq t) \leq \text{Var}(X)/t^2$.*

Proof. Notice that $\mathbb{P}(|X - \mathbb{E}(X)| \geq t) = \mathbb{P}((X - \mathbb{E}(X))^2 \geq t^2)$, so Markov's Inequality implies it is bounded above by $\mathbb{E}((X - \mathbb{E}(X))^2)/t^2 = \text{Var}(X)/t^2$. ■

A particular corollary we will make use of is that

$$\mathbb{P}(X = 0) \leq \mathbb{P}(|X - \mathbb{E}(X)| \geq \mathbb{E}(X)) \leq \frac{\text{Var}(X)}{\mathbb{E}(X)^2}.$$

Another important concept is that of conditional probability.

Definition 1.10. If A and B are events with $\mathbb{P}(B) \neq 0$, the *conditional probability of A given B* is $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$.

Intuitively, we would want A to be independent from B if $\mathbb{P}(A|B) = \mathbb{P}(A)$. Since this requires $\mathbb{P}(B) \neq 0$, we have the slightly more general definition below.

Definition 1.11. Events A and B are said to be *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ (if $\mathbb{P}(B) \neq 0$, this means $\mathbb{P}(A|B) = \mathbb{P}(A)$). More generally, A is *mutually independent* from $B_1, \dots, B_k$ if A is independent of $\bigcap_{j \in J} B_j$ for every $J \subseteq [k]$. We say that $B_1, \dots, B_k$ are *mutually independent* if each B_i is mutually independent from $B_1, \dots, B_{i-1}, B_{i+1}, \dots, B_k$.

Mutual independence is in general stronger than pairwise independence. For example, if we let Ω be the red-blue colorings of the cycle C_5 and A_i be the event that i and $i+1$ are colored the same color, then $\mathbb{P}(A_i) = 1/2$ and $\mathbb{P}(A_i|A_j) = 1/2$ for $i \neq j$, so that the A_i are pairwise independent. However, $\mathbb{P}(A_5|A_1 \cap A_2 \cap A_3 \cap A_4) = 1$, so the A_i are not mutually independent.

Definition 1.12. If $A_1, \dots, A_n \subseteq \Omega$ are events, a *dependency digraph* is a directed graph D on $[n]$ such that each A_i is mutually independent of $\{A_i : (i, j) \notin D\}$.

Notice that K_n is always a dependency digraph. Dependency digraph by no means have to be unique. Often, we will want to find the probability of avoiding some events. To do this, we will use the Lovász Local Lemma.

Theorem 1.13 (Lovász Local Lemma). *Let $A_1, \dots, A_n \subseteq \Omega$ be events and $D = ([n], E)$ a dependency digraph. If there are $x_i \in [0, 1)$ such that $\mathbb{P}(A_i) \leq x_i \prod_{(i,j) \in E} (1 - x_j)$, then $\mathbb{P}(\bigcap_{j=1}^n \overline{A_j}) \geq \prod_{j=1}^n (1 - x_j) > 0$.*

Proof. Notice that

$$\mathbb{P} \left(\bigcap_{j=1}^n \overline{A_j} \right) = \mathbb{P}(\overline{A_1}) \cdot \mathbb{P}(\overline{A_2}|\overline{A_1}) \cdot \mathbb{P}(\overline{A_3}|\overline{A_1} \cap \overline{A_2}) \cdot \dots \cdot \mathbb{P} \left(\overline{A_n} \mid \bigcap_{j=1}^{n-1} \overline{A_j} \right).$$

Thus, it suffices to show that $\mathbb{P}(\overline{A_i} | \bigcap_{j=1}^{i-1} \overline{A_j}) \geq 1 - x_i$. Equivalently, we can show that $\mathbb{P}(A_i | \bigcap_{j=1}^{i-1} \overline{A_j}) \leq x_i$. We will show that, for any $S \subseteq [n]$ and $i \in [n] \setminus S$, $\mathbb{P}(A_i | \bigcap_{j \in S} \overline{A_j}) \leq x_i$.

We proceed by induction on $|S|$. If $|S| = 0$, so that $S = \emptyset$,

$$\mathbb{P}\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) = \mathbb{P}(A_i) = x_i \cdot \prod_{(i,j) \in E} (1 - x_j) \leq x_i$$

for all $i \in [n]$.

Suppose that the statement holds for all $S' \subseteq [n]$ with $|S'| < |S|$. Take $i \in [n] \setminus S$. Write $S_1 = \{j \in S : (i, j) \in E\}$ and $S_2 = S \setminus S_1$. If $S_1 = \emptyset$, then A_i is mutually independent of $\{\overline{A_j} : j \in S_2 = S\}$, so that

$$\mathbb{P}\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) = \mathbb{P}(A_i) = x_i \cdot \prod_{(i,j) \in E} (1 - x_j) \leq x_i.$$

If $S_1 \neq \emptyset$, then

$$\begin{aligned} \mathbb{P}\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) &= \frac{\mathbb{P}\left(A_i \cap \bigcap_{j \in S} \overline{A_j}\right)}{\mathbb{P}\left(\bigcap_{j \in S} \overline{A_j}\right)} = \frac{\mathbb{P}\left(A_i \cap \bigcap_{j \in S_1} \overline{A_j} \cap \bigcap_{\ell \in S_2} \overline{A_\ell}\right)}{\mathbb{P}\left(\bigcap_{j \in S_2} \overline{A_j}\right)} \cdot \frac{\mathbb{P}\left(\bigcap_{j \in S_2} \overline{A_j}\right)}{\mathbb{P}\left(\bigcap_{j \in S_1} \overline{A_j} \cap \bigcap_{\ell \in S_2} \overline{A_\ell}\right)} \\ &= \frac{\mathbb{P}\left(A_i \cap \bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right)}{\mathbb{P}\left(\bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right)}. \end{aligned}$$

We have that

$$\mathbb{P}\left(A_i \cap \bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right) \leq \mathbb{P}\left(A_i \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right) = \mathbb{P}(A_i) \leq x_i \cdot \prod_{(i,j) \in E} (1 - x_j).$$

Writing $S_1 = \{j_1, \dots, j_r\}$,

$$\begin{aligned} \mathbb{P}\left(\bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right) &= \mathbb{P}\left(\overline{A_{j_1}} \cap \dots \cap \overline{A_{j_r}} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right) \\ &= \mathbb{P}\left(\overline{A_{j_1}} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right) \mathbb{P}\left(\overline{A_{j_2}} \middle| \overline{A_{j_1}} \cap \bigcap_{\ell \in S_2} \overline{A_\ell}\right) \dots \mathbb{P}\left(\overline{A_{j_r}} \middle| \overline{A_{j_1}} \cap \dots \cap \overline{A_{j_{r-1}}} \cap \bigcap_{\ell \in S_2} \overline{A_\ell}\right). \end{aligned}$$

Since there are always strictly less than $|S|$ terms in the intersections for the conditional probability, we can use induction to get that this is $\geq \prod_{j \in S_1} (1 - x_j) \geq \prod_{(i,j) \in E} (1 - x_j)$. Thus,

$$\mathbb{P}\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) = \frac{\mathbb{P}\left(A_i \cap \bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right)}{\mathbb{P}\left(\bigcap_{j \in S_1} \overline{A_j} \middle| \bigcap_{\ell \in S_2} \overline{A_\ell}\right)} \leq \frac{x_i \cdot \prod_{(i,j) \in E} (1 - x_j)}{\prod_{(i,j) \in E} (1 - x_j)} \leq x_i,$$

as desired. ■

Corollary 1.14 (Symmetric Lovász Local Lemma). *Let $A_1, \dots, A_n$ be events with $\mathbb{P}(A_i) \leq p$ and let d be the maximal outdegree in the dependency digraph. If $ep(d+1) \leq 1$, then $\mathbb{P}\left(\bigcap_{j=1}^n \overline{A_j}\right) > 0$.*

Proof. Suppose that $ep(d+1) \leq 1$. Notice that we must always have $p < 1$. Now, if $d = 0$, then the A_j are mutually independent, so $\mathbb{P}\left(\bigcap_{j=1}^n \overline{A_j}\right) = \prod_{j=1}^n \mathbb{P}(\overline{A_j}) > 0$. Suppose that $d \geq 1$, and let $x_i = 1/d + 1$ in the LLL. We have that

$$x_i \prod_{(i,j) \in E} (1 - x_j) \geq \frac{1}{d+1} \left(1 - \frac{1}{d+1}\right)^d \geq \frac{1}{d+1} \cdot \frac{1}{e} \geq p \geq \mathbb{P}(A_i),$$

so LLL applies. Thus, $\mathbb{P}\left(\bigcap_{j=1}^n \overline{A_j}\right) > 0$. ■

Independence also comes up in the context of random variables.

Definition 1.15. Random variables X and Y are said to be *independent* if, for all $x, y \in \mathbb{R}$, $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y = y)$. More generally, we say X is *mutually independent* of $Y_1, \dots, Y_k$ if

$$\mathbb{P}(X = x, Y_j = y_j \quad \forall j \in J) = \mathbb{P}(X = x) \cdot \mathbb{P}(Y_j = y_j \quad \forall j \in J)$$

for all $x, y_j \in \mathbb{R}$ and $J \subseteq [k]$.

If X and Y are independent, then the definition of expected value shows that $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$. This in particular means that $\text{Cov}(X, Y) = 0$.

Theorem 1.16 (Chernoff). *If $X_1, \dots, X_n$ are mutually independent random variables with $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = 1/2$, and $X = \sum X_i$, then $\mathbb{P}(X \geq t) \leq e^{-t^2/2n}$.*

Proof. For $\lambda > 0$, $\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t})$. By Markov's Inequality, this is at most $e^{-\lambda t} \mathbb{E}(e^{\lambda X})$. Now, since the X_i are mutually independent,

$$e^{-\lambda t} \mathbb{E}(e^{\lambda X}) = e^{-\lambda t} \mathbb{E}(e^{\lambda \sum X_i}) = e^{-\lambda t} \prod \mathbb{E}(e^{\lambda X_i}) = e^{-\lambda t} \prod \frac{1}{2}(e^\lambda + e^{-\lambda}).$$

It is a fact that $\frac{1}{2}(e^\lambda + e^{-\lambda}) \leq e^{\lambda^2/2}$ (this may be seen, for example, by expanding the Taylor series). Thus,

$$\mathbb{P}(X \geq t) \leq e^{-\lambda t} \prod \frac{1}{2}(e^\lambda + e^{-\lambda}) \leq e^{n\lambda^2/2 - \lambda t}.$$

This is maximized at $\lambda = t/n$, so

$$\mathbb{P}(X \geq t) \leq e^{n\lambda^2/2 - \lambda t} \leq e^{-t^2/2n}.$$

■

This result can be generalized more generally to variables bounded between -1 and 1 :

Theorem 1.17. *Let $X_1, \dots, X_n$ be mutually independent random variables with $|X_i| \leq 1$ and $\mathbb{E}(X_i) = 0$. If $X = \sum X_i$, then $\mathbb{P}(X \geq t) \leq e^{-t^2/2n}$.*

Proof. Let

$$h(x) = \frac{1}{2}(e^\lambda + e^{-\lambda}) + \frac{1}{2}(e^\lambda - e^{-\lambda})x.$$

Expanding the Taylor series shows that $e^{\lambda x} \leq h(x)$ for $|x| \leq 1$. Thus,

$$\mathbb{E}(e^{\lambda X_i}) \leq \mathbb{E}(h(X_i)) = h(\mathbb{E}(X_i)) = \frac{1}{2}(e^\lambda + e^{-\lambda}).$$

Applying the same steps as in the above proof yields the result. ■

1.3 Applying Probabilistic Ideas

We'll apply probabilistic ideas to prove the following result:

Proposition 1.18. *Let G be a graph with m edges. Then G contains a bipartite subgraph with at least $m/2$ edges.*

Proof. Our first instinct would be to randomly include edges into our subgraph, but this may discard edges that can be useful. Because of this, we will instead randomly color vertices and look at the edges between them. Randomly color each vertex red or blue independently with probability $1/2$. For a fixed edge e , let $X_e = \begin{cases} 1 & e \text{ bicolored} \\ 0 & e \text{ monochromatic} \end{cases}$, so that $X = \sum_e X_e$ is the number of edges in our bipartite graph. We have that $\mathbb{E}(X_e) = \mathbb{P}(X_e = 1) = 1/2$, so that $\mathbb{E}(X) = \sum_e \mathbb{E}(X_e) = m/2$. Thus, there must be some coloring with $X \geq m/2$, giving a bipartite subgraph with $\geq m/2$ edges. ■

This bound is almost optimal: if $G = K_n$ and we pick b edges to be blue, then there are $b(n-b) \leq n^2/4$ edges in the bipartite subgraph, while there are $\binom{n}{2} \approx n^2/2$ edges in the whole graph.

Our main idea was that, because $\mathbb{E}(X)$ was $m/2$, there must be *some* coloring that takes on a value at least $m/2$.

1.4 Applying Orthogonality

We will explore the technique of using orthogonality for counting arguments.

Definition 1.19. A family $\mathcal{F}$ of subsets of $[n]$ is an *antichain* if, for all $A \neq B \in \mathcal{F}$, $A \not\subseteq B$ and $B \not\subseteq A$. $\mathcal{F}$ is a *chain* if, for all $A \neq B \in \mathcal{F}$, $A \subseteq B$ or $B \subseteq A$.

Example. If all elements of $\mathcal{F}$ have size k , then $\mathcal{F}$ is an antichain. We have that $|\mathcal{F}| \leq \binom{n}{k} \leq \binom{n}{n/2}$. We will see shortly that this bound is true of all antichains.

Example. If $\pi : [n] \rightarrow [n]$ is a permutation, then

$$\{\emptyset, \{\pi(1)\}, \{\pi(1), \pi(2)\}, \dots, \{\pi(1), \dots, \pi(n)\}\}$$

is a chain.

The idea of orthogonality is to note that a given chain and antichain can intersect in at most one element, and so they are in a sense “orthogonal”. This fact can allow us to deduce information about the size of an antichain.

Theorem 1.20 (LYM Inequality). *Let $\mathcal{F}$ be an antichain of subsets of $[n]$. Then $\sum_{A \in \mathcal{F}} 1/\binom{n}{|A|} \leq 1$.*

Proof. Pick a random maximal chain $\mathcal{C}$ (maximal chains correspond to including each element of $[n]$ one by one in some order, which is naturally in bijection with permutations of $[n]$, so there are $n!$ maximal chains). We know that $|\mathcal{F} \cap \mathcal{C}| \leq 1$.

For $A \in \mathcal{F}$, let $X_A = \begin{cases} 1 & A \in \mathcal{C} \\ 0 & A \notin \mathcal{C} \end{cases}$. We have that $\mathbb{P}(X_A = 1) = \frac{|A|! \cdot (n-|A|)!}{n!} = 1/\binom{n}{|A|}$ (for $\mathcal{C}$ to include A , it

must first include all the elements of A in some order, followed by all the elements of $[n] \setminus A$ in some order). Let $X = \sum_{A \in \mathcal{F}} X_A$ denote the size of $\mathcal{F} \cap \mathcal{C}$. We have that $|\mathcal{F} \cap \mathcal{C}| \leq 1$, so $X \leq 1$. Taking expectations,

$$1 \geq \mathbb{E}(X) = \sum_{A \in \mathcal{F}} \mathbb{E}(X_A) = \sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}}.$$

■

Corollary 1.21 (Sperner’s Theorem). *Let $\mathcal{F}$ be an antichain of subsets of $[n]$. Then $|\mathcal{F}| \leq \binom{n}{n/2}$.*

Proof. For any $A \subseteq [n]$, $\binom{n}{|A|} \leq \binom{n}{n/2}$. By the LYM inequality,

$$1 \geq \sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{|A|}} \geq \sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{n/2}} = \frac{|\mathcal{F}|}{\binom{n}{n/2}},$$

so $|\mathcal{F}| \leq \binom{n}{n/2}$.

■

This idea of picking a random orthogonal element and examining intersection size is very useful for proving bounds.

Definition 1.22. A family $\mathcal{F}$ of subsets of $[n]$ is said to be *intersecting* if, for all $A, B \in \mathcal{F}$, $A \cap B \neq \emptyset$.

Example. The family $\{A \cup \{n\} : A \subseteq [n-1]\}$ is intersecting. For $k > n/2$, the family $\binom{[n]}{k}$ is intersecting.

To prove a bound on the maximum size of intersecting sets, we will once again look for orthogonality.

Lemma 1.23. *Let $n \geq 2k$, and define $A_s = \{s, s+1, \dots, s+k-1\} \subseteq \mathbb{Z}/n\mathbb{Z}$. Then the largest intersecting subfamily of $\{A_s : s \in \mathbb{Z}/n\mathbb{Z}\}$ has size k .*

Proof. Let $H \subseteq \{A_s : s \in \mathbb{Z}/n\mathbb{Z}\}$ be intersecting, and fix $A_s \in H$. Then A_s intersects $A_{s-k+1}, \dots, A_{s+k-1}$. Pair these as $(A_{s-k+1}, A_{s+1}), (A_{s-k+2}, A_{s+2}), \dots, (A_{s-1}, A_{s+k-1})$. These pairs are disjoint (since $n \geq 2k$), so H contains at most set from each pair. Thus, $|H| \leq k$.

■

Now, for a general intersecting set, we will have to consider subsets of $\mathbb{Z}/n\mathbb{Z}$ that are not consecutive. The workaround to this problem is to consider random permutations applied to the sets A_s to get a random k element set.

Theorem 1.24 (Erdős-Ko-Rado). *Let $n \geq 2k$ and let H be an intersecting k -uniform hypergraph on $[n]$. Then $|H| \leq \binom{n-1}{k-1}$.*

Proof. Let H be an intersecting k -uniform hypergraph. For a fixed permutation $\pi \in S_n$ and a randomly chosen $s \in \mathbb{Z}/n\mathbb{Z}$, we have that $\mathbb{P}_{s \in \mathbb{Z}/n\mathbb{Z}}(\pi(A_s) \in H) \leq k/n$, as H can contain at most k of the sets $\pi(A_s)$.

Randomly pick a pair $(\pi, s) \in S_n \times \mathbb{Z}/n\mathbb{Z}$. Then $\pi(A_s)$ gives a k -element subset of $\mathbb{Z}/n\mathbb{Z}$ uniformly at random, so that $\mathbb{P}_{\pi \in S_n, s \in \mathbb{Z}/n\mathbb{Z}}(\pi(A_s) \in H) = |H|/\binom{n}{k}$. Now,

$$|H| = \binom{n}{k} \mathbb{P}_{\pi \in S_n, s \in \mathbb{Z}/n\mathbb{Z}}(\pi(A_s) \in H) = \binom{n}{k} \cdot \frac{1}{n!} \sum_{\pi \in S_n} \mathbb{P}_{s \in \mathbb{Z}/n\mathbb{Z}}(\pi(A_s) \in H) \leq \binom{n}{k} \cdot \frac{1}{n!} \cdot n! \cdot \frac{k}{n} = \binom{n-1}{k-1}.$$

■

There are many good ideas in this proof:

- Orthogonality: we are able to bound $|H \cap \pi(A_s)|$ and take advantage of that bound
- Randomly permuting things that we understand to get things that we don't
- Averaging over randomly chosen elements to take advantage of bounds for fixed elements

1.5 Applying Alterations

The alterations technique involves finding an object that has nice properties and then modifying it to have the properties that we want. We will demonstrate this technique by constructing a graph on $[n]$ that avoids having $K_{3,3}$ as a subgraph.

For the purpose of this construction, we will mostly focus on asymptotics. Let $G(n, p)$ be the Erdős-Rényi random graph where we include each edge independently with probability p . Define the random variable X to be the number of edges in the graph, and Y to be the number of $K_{3,3}$ subgraphs. We can write $X = \sum_{e \in \binom{[n]}{2}} X_e$, where

$$X_e = \begin{cases} 1 & e \text{ present} \\ 0 & \text{else} \end{cases}. \text{ Then,}$$

$$\mathbb{E}(X) = \sum_{e \in \binom{[n]}{2}} \mathbb{E}(X_e) = p \binom{n}{2}.$$

Similarly, we can write $Y = \sum_{A, B \in \binom{[n]}{3}} Y_{A,B}$, where $Y_{A,B} = \begin{cases} 1 & \text{all edges } A-B \text{ present} \\ 0 & \text{else} \end{cases}$. We have that

$$\mathbb{E}(Y) = \sum_{A, B \in \binom{[n]}{3}} \mathbb{E}(Y_{A,B}) = p^9 \binom{n}{3} \binom{n-3}{3}.$$

Since we only care about asymptotics, we can ignore factors of 2 that may show up. Now,

$$\mathbb{E}(X - Y) = p \binom{n}{2} - p^9 \binom{n}{3} \binom{n-3}{3} \geq \frac{1}{3}pn^2 - \frac{1}{36}p^9n^6,$$

where we recall that we only care about large n . Taking $p = n^{-1/2}$ (as this means $n^2p = n^6p^9$), we have that $\mathbb{E}(X - Y) \geq \frac{11}{36}n^{3/2}$. Now, there is some graph G with $(X - Y)(G) \geq \frac{11}{36}n^{3/2}$. Delete 1 edge in every $K_{3,3}$ subgraph of G to obtain a new graph with $(X - Y)(G) \geq \frac{11}{36}n^{3/2}$ edges. Thus, asymptotically, we can construct a graph on $[n]$ with $n^{3/2}$ edges that contains no $K_{3,3}$ subgraphs.

This technique of alterations is very powerful.

Definition 1.25. If G is a graph, let $\alpha(G)$ denote the size of a largest independent set of G .

Proposition 1.26. *Let G be a graph with n vertices and $\leq nd/2$ edges for some $d \geq 1$. Then $\alpha(G) \geq n/2d$.*

Proof. Let S be a random subset of vertices, where S contains a given vertex independently with probability p . Let the random variable X be the size of S and the random variable Y be the number of edges with endpoints in S . We see that $\mathbb{E}(X) = np$ and $\mathbb{E}(Y) \leq \frac{nd}{2}p^2$, so that $\mathbb{E}(X - Y) \geq np - \frac{nd}{2}p^2$. Taking $p = 1/d$ (which maximizes the expected value), $\mathbb{E}(X - Y) = n/2d$. Thus, take some set S with $(X - Y)(S) \geq n/2d$ and delete one vertex from every edge with endpoints in S to get an independent set S' with $|S'| \geq n/2d$. ■

Definition 1.27. If G is a graph, the *girth* $g(G)$ is the length of a shortest cycle. The *chromatic number* $\chi(G)$ is the smallest number of colors needed to color G .

We would expect graphs of large girth to have a low chromatic number, as large cycles do not obstruct colorings that much. However, this is false.

Theorem 1.28 (Erdős). *For every $k, \ell \geq 0$, there is a graph G with $g(G) \geq \ell$ and $\chi(G) \geq k$.*

Proof. Let $G = G(n, p)$ be an Erdős-Rényi random graph, with n large enough that we can perform asymptotics. If X_t denotes the number of cycles of length t and $X = \sum_{t=3}^{\ell-1} X_t$ is the number of cycles of length $\leq \ell - 1$, then

$$\mathbb{E}(X) = \sum_{t=3}^{\ell-1} \mathbb{E}(X_t) = \sum_{t=3}^{\ell-1} \binom{n}{t} \cdot \frac{t!}{2t} \cdot p^t = \sum_{t=3}^{\ell-1} n^t p^t.$$

Let $p = n^{-1+\varepsilon}$, so that $\mathbb{E}(X) = \sum_{t=3}^{\ell-1} n^t p^t = \sum_{t=3}^{\ell-1} n^{t\varepsilon}$. Taking $\varepsilon = 1/2\ell$, we see that $\mathbb{E}(X) \in o(n)$ (that is, $\lim_{n \rightarrow \infty} \mathbb{E}(X)/n = 0$). Thus, if we take n large enough that $\mathbb{E}(X) < n/4$, Markov's inequality implies $\mathbb{P}(X \geq n/2) < 1/2$. This gives us a way to control the girth of the graph.

Now, we want to control the chromatic number. Notice that colorings of graphs are the same as partitions into independent sets, so that $\chi(G)\alpha(G) \geq n$. Thus, we can instead study $\alpha(G)$. We have that

$$\mathbb{P}(\alpha(G) \geq a) \leq \binom{n}{a} (1-p)^{\binom{a}{2}} \leq n^a e^{-p \binom{a}{2}} = e^{a(\log n - p \cdot \frac{a-1}{2})}.$$

Taking $a = \frac{3}{p} \log n + 1$, we see that $\mathbb{P}(\alpha(G) \geq a)$ goes to 0 as n goes to ∞ .

We now put these together. Take n to be sufficiently large that $\mathbb{E}(X) < n/4$ (so that $\mathbb{P}(X \geq n/2) < 1/2$) and $\mathbb{P}(\alpha(G) \geq a) < 1/2$. Additionally, let n be large enough that $n/a \geq k$ (which is possible as n/a tends to ∞ with n). Since both events $\mathbb{P}(X \geq n/2)$ and $\mathbb{P}(\alpha(G) \geq a)$ have probability less than half, there must be some graph where neither event occurs. Let G be such a graph, so that $X(G) \leq n/2$ and $\alpha(G) < a$. Delete one vertex from each cycle of G of length $\leq \ell - 1$, so that we get a graph G' with $g(G') \geq \ell$. Notice that we cannot possibly increase the size of an independent set through this process, so $\alpha(G') < a$. Now, $\chi(G') \geq n/\alpha(G') > n/a \geq k$, so G' is our desired graph. ■

1.6 Applying the Second Moment Method

Suppose that $G = G(n, p)$, and let the random variable X be the number of triangles in G . We see that $\mathbb{E}(X) = \binom{n}{3} p^3 \leq n^3 p^3$. If $np \rightarrow 0$ as $n \rightarrow \infty$, then $\mathbb{E}(X) \rightarrow 0$. By Markov's Inequality, $\mathbb{P}(X > 0) = \mathbb{P}(X \geq 1) \leq \mathbb{E}(X) \rightarrow 0$, meaning that $\mathbb{P}(X = 0) \rightarrow 1$. However, if $np \rightarrow \infty$, this doesn't necessarily imply $\mathbb{P}(X > 0) \rightarrow 1$. However, this will be the case if X is typically close to its expected value. In other words, to show a result of this form, we'll want to look at the variance.

Suppose $np \rightarrow \infty$ as $n \rightarrow \infty$. We can write $X = \sum_{\tau \in \binom{[n]}{3}} X_\tau$, where X_τ is 1 if the vertices of τ form a triangle. Notice that $\text{Var}(X_\tau) = p^3 - p^6 \leq p^3$. If τ and τ' are disjoint, then $\mathbb{P}(X_\tau = X_{\tau'} = 1) = p^6$, so $\text{Cov}(X_\tau, X_{\tau'}) = p^6 - p^6 = 0$. If τ and τ' share an edge, then $\mathbb{P}(X_\tau = X_{\tau'} = 1) = p^5$, so $\text{Cov}(X_\tau, X_{\tau'}) = p^5 - p^6 \leq p^5$. Thus,

$$\mathbb{P}(X = 0) \leq \frac{\text{Var}(X)}{\mathbb{E}(X)^2} = \frac{\sum_{\tau} \text{Var}(X_\tau) + \sum_{\tau, \tau'} \text{Cov}(X_\tau, X_{\tau'})}{\mathbb{E}(X)^2} \leq \frac{n^3 p^3 + 2n^4 p^5}{Cn^6 p^6} \xrightarrow{n \rightarrow \infty} 0.$$

Thus, $\mathbb{P}(X \geq 1) \rightarrow 1$.

This is the threshold phenomenon: if $p \ll 1/n$ (i.e. $p \in o(1/n)$), then the probability of triangles appearing goes to 0. On the other hand, if $p \gg 1/n$, then the probability triangles appearing goes to 1. We can generalize this phenomenon.

Definition 1.29. If G is a graph with n vertices and m edges, the *density* of G is $\delta(G) = m/n$.

Definition 1.30. A graph G is said to be *balanced* if no subgraph of G has a higher density than G .

Note that triangles are balanced, so what we have established is a special case of the following result.

Theorem 1.31. *Let H be a balanced graph. Then the threshold for the appearance of H as a subgraph of $G(n, p)$ is $n^{-1/\delta(H)}$.*

Proof. Let H have vertex set $[v]$ and have e edges. Let X be the number of appearances of H as a subgraph of $G(n, p)$. If $\alpha = (a_1, \dots, a_v)$ is an ordered tuple of distinct vertices in $G \sim G(n, p)$, define

$$X_\alpha = \begin{cases} 1 & (i, j) \in E(H) \implies (\alpha_i, \alpha_j) \in E(G) \\ 0 & \text{else} \end{cases},$$

so that $X = \sum_\alpha X_\alpha$. We have that $\mathbb{E}(X) = \sum_\alpha \mathbb{E}(X_\alpha) \leq n^v p^e$. If $p \ll n^{-1/\delta(H)} = n^{-v/e}$, then this goes to 0 as $n \rightarrow \infty$. Thus, $\mathbb{P}(X \geq 1) \leq \mathbb{E}(X) \xrightarrow{n \rightarrow \infty} 0$.

Now, $\text{Var}(X) = \sum_{\alpha, \beta} \text{Cov}(X_\alpha, X_\beta)$. If α and β share ≤ 1 vertex, the covariance is 0. If they share $t \geq 2$ vertices, then they share $\leq t\delta$ edges (since H is balanced), meaning there are a total of $\geq 2e - t\delta$ edges between them. Thus, $\mathbb{E}(X_\alpha X_\beta) \leq p^{2e-t\delta}$, so $\text{Cov}(X_\alpha, X_\beta) \leq p^{2e-t\delta}$. There are on the order of n^{2v-t} pairs that share t vertices, so

$$\text{Var}(X) \leq \sum_{t=2}^v C n^{2v-t} p^{2e-t\delta} \leq C \sum_{t=2}^v (n^v p^e)^{2-t/v}.$$

Since $\mathbb{E}(X) \geq C' n^v p^e$, we have that

$$\mathbb{P}(X = 0) \leq \frac{\text{Var}(X)}{\mathbb{E}(X)^2} \leq \frac{C}{C'^2} \sum_{t=2}^v (n^v p^e)^{-t/v}.$$

If $p \gg n^{-1/\delta(H)} = n^{-v/e}$, then this goes to 0 as $n \rightarrow \infty$. Thus, $\mathbb{P}(X = 0) \xrightarrow{n \rightarrow \infty} 0$. ■

We'll say $\{x_1, \dots, x_k\} \subseteq \mathbb{Z}_{>0}$ has distinct sums if $\sum_{i \in I} x_i = \sum_{j \in J} x_j$ implies $I = J$ for all $I, J \subseteq [k]$. For example, $\{2^0, 2^1, \dots, 2^{k-1}\}$ has distinct sums. We'll let $f(n)$ denote the maximum k for which there is a set $\{x_1, \dots, x_k\} \subseteq [n]$ with distinct sums.

Let $\varepsilon_1, \dots, \varepsilon_k$ be independent random variables with $\mathbb{P}(\varepsilon_i = 0) = \mathbb{P}(\varepsilon_i = 1) = 1/2$. Let $X = \sum_{i=1}^k \varepsilon_i x_i$. We see that $\mathbb{E}(X) = \frac{1}{2} \sum_{i=1}^k x_i$ and $\text{Var}(X) = \frac{1}{4} \sum_{i=1}^k x_i^2 \leq kn^2/4$. If $\sigma = \sqrt{\text{Var}(X)} \leq n\sqrt{k}/2$, then $\mathbb{P}(|X - \mathbb{E}(X)| \geq t\sigma) \leq \text{Var}(X)/t^2\sigma^2 = 1/t^2$. Thus, $1 - 1/t^2 \leq \mathbb{P}(|X - \mathbb{E}(X)| < t\sigma)$. Additionally, there are $2t\sigma + 1$ possible values of X with $|X - \mathbb{E}(X)| < t\sigma$, and the property of distinct sums means each of these occurs with probability at most $1/2^k$. thus, $\mathbb{P}(|X - \mathbb{E}(X)| < t\sigma) \leq (2t\sigma + 1)/2^k$, so $1 - 1/t^2 \leq (2t\sigma + 1)/2^k \leq (nt\sqrt{k} + 1)/2^k$. Thus, $n \geq \frac{1}{t\sqrt{k}} (2^k(1 - 1/t^2) - 1)$ for all t . We can see that, for large n , $k = f(n) \leq \log_2(n) + \frac{1}{2} \log_2 \log_2(n) + O(1)$.

1.7 Applying the Lovász Local Lemma

Suppose that we have a partition of the n -cycle C_n into c color classes of size k . We want to find a rainbow independent set. We can randomly construct a rainbow set by uniformly at random picking one vertex of each color. There are n bad events: namely, the events A_i where we select both i and $i + 1$. We therefore want $\mathbb{P}(\bigcap_{i=1}^n \overline{A_i}) > 0$.

We can see that $\mathbb{P}(A_i) \leq 1/k^2$, so that $\mathbb{P}(\bigcup_{i=1}^n A_i) \leq n/k^2 = c/k$. Thus, if $k > c$, we are done. However, we'd like to find a constant bound for k , rather than one that depends on c . Doing this requires applying the LLL.

A_i only depends on the A_j where at least one of j and $j + 1$ is colored the same as i or $i + 1$. There are $\leq 2k - 1$ possibilities for one of j and $j + 1$ to be colored the same as i (the $2(k - 1)$ edges where the vertex that is colored the same as i is not i , along with the edge $(i - 1, i)$), and $\leq 2k - 1$ possibilities for one of j and $j + 1$ to be colored the same as $i + 1$. Thus, $d \leq 4k - 2$. If the Symmetric LLL applies, we are done. We need $ep(d + 1) \leq 1$, so that $e(4k - 1)/k^2 \leq 1$. In the integers, this happens when $k \geq 11$.

Theorem 1.32. *Let H be a k -uniform hypergraph such that each edge intersects $\leq d$ edges. If $e(d + 1) \leq 2^{k-1}$, then H is 2-colorable.*

Proof. Color each vertex red or blue uniformly. The bad events are $A_f = \{f \text{ monochromatic}\}$ for edges f , so that $\mathbb{P}(A_f) = 1/2^{k-1}$. A_f is independent from all the edges that f does not intersect, so the outdegree of f in the dependency digraph is at most d . Thus, since $e(d + 1) \leq 2^{k-1}$, the Symmetric LLL applies, meaning that $\mathbb{P}(\bigcap \overline{A_f}) > 0$, so H is 2-colorable. ■

1.8 Applying Compactness

The idea of compactness is useful for extending results about finite subsets to results about infinite subsets.

Definition 1.33. Given a coloring $c : X \rightarrow [k]$, we say $A \subseteq X$ is *rainbow* if $c(A) = [k]$.

Theorem 1.34. *Let $k \geq 1$. Then there is some $m \in \mathbb{N}$ such that, for any $S \subseteq \mathbb{R}$ with $|S| = m$, there is a coloring $c : \mathbb{R} \rightarrow [k]$ such that $x + S$ is rainbow for all $x \in \mathbb{R}$.*

We can prove a finite version of this:

Proposition 1.35. *Let $k \geq 1$. Then there is some $m \in \mathbb{N}$ such that, for any $S, X \subseteq \mathbb{R}$ with $|S| = m$ and X finite, there is a coloring $c : \mathbb{R} \rightarrow S$ such that $x + S$ is rainbow for every $x \in X$.*

Proof. We only care about how a coloring looks on $\bigcup_{x \in X} x + S$. Take a random coloring c of $\bigcup_{x \in X} x + S$ (coloring each element randomly and uniformly), and let A_x be the event that $x + S$ is not rainbow under c . There are at most $k(k-1)^m$ colorings such that $x + S$ is not rainbow (pick which element of $[k]$ is not used to color $x + S$), so $\mathbb{P}(A_x) \leq k(1-1/k)^m$. Now, A_x is independent of all A_y such that $(x+S) \cap (y+S) = \emptyset$, and each $x + S$ intersects at most m^2 other translates of S (as there are m^2 choices for which elements of $x + S$ and $y + S$ are equal). Thus, if we take m large enough that $e(m^2+1)k(1-1/k)^m \leq 1$, we may apply the Symmetric LLL to get a coloring c such that every translate of S is rainbow. ■

Now, we want a coloring, which is an element of $[k]^{\mathbb{R}}$. Put the discrete topology on $[k]$ and the product topology on $[k]^{\mathbb{R}}$. Since $[k]$ is compact, Tychonoff's Theorem means that $[k]^{\mathbb{R}}$ is compact. For $x \in \mathbb{R}$, we'll let $C_x = \{c \in [k]^{\mathbb{R}} : c(x+S) = [k]\}$. Since C_x only depends on finitely many coordinates, it is closed in the product topology. The above result means that any finite collection $\{C_x : x \in X\}$ intersects nontrivially, so compactness means that $\{C_x : x \in \mathbb{R}\}$ intersects nontrivially (if every finite collection of closed sets intersects nontrivially, then every finite collection of their complements does not form an open cover, so the collection of all their complements does not form an open cover, so they all intersect nontrivially).

1.9 Applying Chernoff Bounds

For a fixed $\delta > 0$, suppose that we want to find a family $\mathcal{F}$ of subsets of $[n]$ that are pairwise very δ -different: for $A, B \in \mathcal{F}$, $|A \Delta B| \geq \delta n$ (recall that $A \Delta B$ denotes the symmetric difference $(A \setminus B) \cup (B \setminus A)$). Equivalently, suppose that we want to find vertices of the n -cube that are pairwise far apart (in the sense of the Hamming distance). If X and Y are randomly chosen points in $\{0, 1\}^n$ and $Z_i = |X_i - Y_i|$ is 1 if they differ in the i th coordinate and 0 if they don't, then $d_H(X, Y) = \sum Z_i$. Additionally, the Z_i are mutually independent and $\mathbb{P}(Z_i = 0) = \mathbb{P}(Z_i = 1) = 1/2$. Thus,

$$\mathbb{P}(d_H(X, Y) \leq \delta n) = \mathbb{P}(\sum Z_i \leq \delta n) = \mathbb{P}(\sum (1 - 2Z_i) \geq (1 - 2\delta)n) \leq e^{-\frac{(1-2\delta)^2 n}{2}}.$$

Thus, the union bound implies that we can pick exponentially many points (in n) on the hypercube $\{0, 1\}^n$ and have a positive probability that they are pairwise at distance $\geq \delta n$.

Suppose that G is a graph such that all vertices have degree 1000. We'll show the existence of a subset U of vertices such that all vertices in $G[U]$ have degree between 100 and 900 (where $G[U]$ is the graph with vertex set U , with an edge between two elements of U if and only if there is an edge between them in G). We'll pick U by randomly including each vertex with probability $1/2$. For each $v \in U$, let A_v be the event that $d_{G[U]}(v) \notin [100, 900]$. Notice that $d_{G[U]}(v)$ is the sum of 999 mutually independent random indicator variables, so Chernoff bounds imply

$$\mathbb{P}(|d_{G[U]}(v) - 500| \geq 400) \leq 2e^{-400^2/2 \cdot 1000},$$

which is a tiny number (in particular, it's less than $1/10^7$). To apply LLL, note that A_v and A_w are not mutually independent if and only if A_v and A_w are neighbors or share a common neighbor, so that $d \leq 10^6 + 10^3 \leq 2 \cdot 10^6$. Now, $e(d+1)p \leq 1$, so LLL applies, and none of the A_v occur with positive probability.

1.10 Martingales

Suppose we wanted to compute $\mathbb{E}(\chi(G(3, 1/2)))$. We could fix an edge and look at the cases when it is and is not included:

$$\mathbb{E}(\chi(G(3, 1/2))) = \frac{1}{2} \mathbb{E} \left(\chi \left(\begin{array}{c} \bullet \\ \diagup \quad \diagdown \\ \bullet \quad \bullet \end{array} \right) \right) + \frac{1}{2} \mathbb{E} \left(\chi \left(\begin{array}{c} \bullet \\ \textcolor{red}{\diagup} \quad \diagdown \\ \bullet \quad \bullet \end{array} \right) \right).$$

We could similarly then look at another edge, and compute

$$\mathbb{E} \left(\chi \left(\begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right) \right) = \frac{1}{2} \mathbb{E} \left(\chi \left(\begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right) \right) + \frac{1}{2} \mathbb{E} \left(\chi \left(\begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \end{array} \right) \right),$$

and continue this process. This idea of successive averaging is called a martingale.

Definition 1.36. If $(\Omega, \mathbb{P})$ is a probability space and $X_0, \dots, X_n : \Omega \rightarrow \mathbb{R}$ are random variables, we say $X_0, \dots, X_n$ is a *martingale* if we can obtain X_{i-1} from X_i in the following way: partition $\Omega = A_1 \sqcup \dots \sqcup A_k$ such that X_{i-1} is constant on A_j and is equal to the average of X_i over A_j .

In our example, $X_0 = \mathbb{E}(\chi(G(3, 1/2)))$, $X_n(G) = \chi(G)$ for $G \sim G(3, 1/2)$, and the X_i are obtained by successively averaging upwards as we described.

Proposition 1.37 (Azuma's Inequality). *If $X_0, \dots, X_n$ is a martingale with $X_0 = \mathbb{E}(X_n)$ and $|X_i - X_{i-1}| \leq 1$, then*

$$\mathbb{P}(|X_n - \mathbb{E}(X_n)| \geq t) \leq e^{-t^2/2n}.$$

This tells us that random variables (such as the chromatic number) that we can create a martingale for are highly concentrated around their expected value.

2 Ramsey Theory

2.1 Ramsey Numbers

Recall that if G is a graph, $\alpha(G)$ denotes the size of a largest independent subset of G , and $\omega(G)$ denotes the size of a largest clique of G .

Definition 2.1. The *Ramsey number* $R(s, t)$ is the minimal n such that every graph G on n vertices has $\alpha(G) \geq s$ or $\omega(G) \geq t$. Equivalently, every red-blue edge coloring of K_n either contains a red copy of K_s or a blue copy of K_t .

We see that $R(s, t) = R(t, s)$, since the coloring definition of $R(s, t)$ is symmetric (equivalently, given a graph on n vertices with $\alpha \geq s$ or $\omega \geq t$, toggling which edges are included gives a graph on n vertices with $\alpha \geq t$ or $\omega \geq s$).

A standard result is the following.

Proposition 2.2. $R(3, 3) = 6$.

Proof. The cycle C_5 contains no cliques or independent sets of size 3, so $R(3, 3) \geq 6$. Consider an edge coloring of K_6 . Fix some vertex v . Then $\deg v = 5$, so at least 3 of the edges connected to v are the same color (WLOG, red). Call these edges $\{v, w_1\}, \{v, w_2\}, \{v, w_3\}$. If any of the edges between the w_i is red, we have found a red copy of K_3 . Otherwise, each of the edges between the w_i is blue, so we have a blue K_3 . ■

Proposition 2.3. $R(t, t) \geq 2^{t/2}$ for $t \geq 3$.

Proof. Let $n = 2^{t/2}$, and color each edge in K_n red or blue independently with probability $1/2$. The probability that a set T of t vertices is monochromatic is $2^{-\binom{t}{2}+1}$, so the probability that there exists a monochromatic copy of K_t is at most

$$\binom{n}{t} 2^{-\binom{t}{2}+1} \leq \frac{1}{t!} n^t 2^{-\binom{t}{2}+1} = \frac{1}{t!} 2^{1+t/2} < 1.$$

Thus, there is some red-blue coloring of K_n containing no monochromatic copies of K_t . ■

To show that Ramsey numbers do exist, note that it suffices to show that $R(t, t)$ is finite (if $s \leq t$, then $R(s, t) \leq R(t, t)$).

Theorem 2.4. $R(t, t) \leq 2^{2t}$.

Proof. Consider a red-blue edge coloring of $K_{2^{2t}}$. At least half of the edges incident to $v_1 = 1$ are the same color, so we can take $S_1 \subseteq [2^{2t}]$ with $|S_1| \geq 2^{2t-1}$ such that every edge from 1 to S_1 has the same color x_1 . Let $v_2 = \min S_1$, and take $S_2 \subseteq S_1$ such that $|S_2| \geq 2^{2t-2}$ and every edge from v_2 to S_2 has the same color x_2 . Inductively construct $v_1, \dots, v_{2t}$ and $S_1, \dots, S_{2t}$ such that edges from v_i to S_i are colored x_i . Now, there is some $I = \{i_1, \dots, i_t\} \subseteq [2t]$ such that the x_{i_j} are the same color. All edges between $\{v_{i_j} : 1 \leq j \leq t\}$ are the same color, so there is a monochromatic copy of K_t . ■

We can extend the proof of this theorem to an infinite analog (Our proof only gets easier, as each $|S_i|$ will be infinite):

Theorem 2.5. *For any red-blue coloring of $\binom{\mathbb{N}}{2}$, there is an infinite $I \subseteq \mathbb{N}$ such that $\binom{I}{2}$ is monochromatic.*

More generally, we have a hypergraph version of this result.

Theorem 2.6. *For any red-blue coloring of $\binom{\mathbb{N}}{k}$, there is an infinite $S \subseteq \mathbb{N}$ such that $\binom{S}{k}$ is monochromatic.*

Proof. We induct on k . When $k = 1$, this is trivial. If we have a coloring $c : \binom{\mathbb{N}}{k} \rightarrow [2]$, define $c'_1 : \binom{\mathbb{N} \setminus \{1\}}{k-1} \rightarrow [2]$ by $A \mapsto c(A \cup \{1\})$. By induction, there is an infinite $S_1 \subseteq \mathbb{N} \setminus \{1\}$ such that all elements of $\binom{S_1}{k-1}$ have the same color $x_1 \in [2]$ under c'_1 . Let $v_2 = \min(S_1)$, and define $c'_2 : \binom{S_1 \setminus \{v_2\}}{k-2} \rightarrow [2]$ by $A \mapsto c(A \cup \{v_2\})$. There is an infinite $S_2 \subseteq S_1 \setminus \{v_2\}$ such that all elements of $\binom{S_2}{k-2}$ have the same color x_2 under c'_2 . Do this inductively to get $v_1, v_2, \dots \in \mathbb{N}$, $\mathbb{N} \supsetneq S_1 \supsetneq S_2 \supsetneq \dots$ such that $c(A \cup \{v_i\}) = x_i$ for $A \in \binom{S_i}{k-1}$. Let $x_{i_1}, x_{i_2}, \dots$ be an infinite constant subsequence. Then any k -element subset of $\{v_{i_1}, v_{i_2}, \dots\}$ has the same color x_{i_1} , as desired. ■

Once again, there is also a finite version.

Theorem 2.7 (Hypergraph Ramsey). *Let $k \geq 2$ and $t \in \mathbb{N}$. Then there is $n \in \mathbb{N}$ such that, for any 2-coloring $\binom{[n]}{k} \rightarrow [2]$, there is $S \subseteq [n]$ of size t such that $\binom{S}{k}$ is monochromatic.*

Proof. We could present a modification of the proof for the infinite case following the ideas from the proof of $R(t, t) \leq 2^{2t}$ (in particular showing that $R_k(t, t) \leq 2^{R_{k-1}(t, t)}$). We instead present a proof knowing that the infinite case is true.

Suppose instead that for any n , there is $c_n : \binom{[n]}{k} \rightarrow [2]$ such that no t -element subset $S \subseteq [n]$ has $\binom{S}{k}$ monochromatic. There are 2 ways to color $\binom{[k]}{k}$, so there is some infinite subsequence $c_{n_1}, c_{n_2}, \dots$ that all agree on $\binom{[k]}{k}$, say with a fixed coloring d_k . Now, there are finitely many colorings of $\binom{[k+1]}{k}$, so we can once again get an infinite subsequence $c_{n_1(2)}, c_{n_2(2)}, \dots$ that all agree with some d_{k+1} on $\binom{[k+1]}{k}$. Inductively, we get $d_k, d_{k+1}, \dots$ with $d_n : \binom{[n]}{k} \rightarrow [2]$ such that d_n has no monochromatic size t subset and d_i and d_{i+1} agree on $\binom{[i]}{k}$. Define $d : \binom{\mathbb{N}}{k} \rightarrow [2]$ by $A \subseteq [n] \mapsto d_n(A)$ (which is well-defined because d_i and d_j agree whenever they are both defined). This avoids monochromatic sets of size t , and so in particular avoids infinite monochromatic sets. This is a contradiction. ■

2.2 Points in the Plane

Recall that we say $X \subseteq \mathbb{R}^d$ are in general position if no $k+2$ points of X lie in a k -dimensional hyperplane for $0 \leq k < d$. We say that X is in convex position if no $x \in X$ lies in the convex hull of $X \setminus \{x\}$, where the convex hull of $Y \subseteq \mathbb{R}^d$ is

$$\text{conv}(Y) = \left\{ \sum_{i=1}^n \lambda_i y_i : y_i \in Y, \lambda_i \geq 0, \sum_{i=1}^n \lambda_i = 1 \right\}.$$

Notice that any 3 points in general position are automatically in convex position. It is possible for 4 points in general position not to be in convex position, as in the configuration below



Notice, however, that wherever we add a fifth point, we must get 4 points in convex position. This is true in general: any 5 points in general position must contain 4 points in convex position. To see this, recall that K_5 is nonplanar, so if we have 5 points in general position, we may draw the straight line edges between them and find a pair that intersect. The endpoints of these edges must then be in convex position. A similar phenomenon is true in general.

Theorem 2.8 (Erdős-Szekeres). *For any k , there is an n such that, for any $X \subseteq \mathbb{R}^2$ in general position with $|X| = n$, there is $Y \subseteq X$ in convex position with $|Y| = k$.*

Proof. This is trivial for $k = 1, 2, 3$, and we have verified it for $k = 4$, so suppose $k \geq 5$. Color $A \in \binom{[n]}{4}$ blue if A is in convex position, and red otherwise. By hypergraph Ramsey, there is a monochromatic $k \geq 5$ -clique for sufficiently large n . By what we established above, a 5-clique cannot be red, so it must be blue. This means that there is a collection of k points such that every 4-tuple is in convex position, which means that all of the k points must be in convex position. ■

2.3 Monochromatic Patterns

We'll call an arithmetic progression of length m an " m -AP".

Definition 2.9. We say that APs $\{a_i, a_i + d_i, \dots, a_i + (m-1)d_i\}$ for $i \in [\ell]$ are *focused at f* if $f = a_i + md_i$ for all i . If we have some coloring of $\mathbb{N}$, we say that they are *color-focused at f* if they are focused at f , each $\{a_i, a_i + d_i, \dots, a_i + (m-1)d_i\}$ is monochromatic, and the colors are pairwise distinct.

Proposition 2.10. *For any k , there exists an n such that any k -coloring of $[n]$ has a monochromatic 3-AP.*

Proof. We have the following claim: for all $r \leq k$, there is an n such that, whenever $[n]$ is k -colored, there is a monochromatic 3-AP or r color-focused 2-APs. Taking $r = k$ implies the result.

We prove the claim by induction on r . For $r = 1$, this is true with $n = t + 1$ (there must be a 2-AP by the pigeonhole principle). Suppose that the claim is true for $r - 1$ with n , and consider a k -coloring of $[2n(k^{2n} + 1)]$. Let B_i be the block $\{2n(i-1) + 1, \dots, 2ni\}$ of length $2n$. There are k^{2n} ways of coloring such a block, and $k^{2n} + 1$ blocks, so there are 2 blocks colored identically, say B_s and B_{s+t} . Now, by induction, there are $r - 1$ color-focused 2-APs $\{a_i, a_i + d_i\}$ in $\{2n(s-1) + 1, \dots, 2n(s-1) + n\} \subseteq B_s$, and their focus f must also lie in B_s (since B_s has size $2n$). Because B_s and B_{s+t} are colored identically, $\{a_i, a_i + d_i + 2tn\}$ is color-focused at $a_i + 2d_i + 4tn = f + 4tn$. Now, if f is colored the same as some $\{a_i, a_i + d_i\}$, we have found a monochromatic 3-AP. Otherwise, $\{f, f + 2tn\}$ is also focused at $f + 4tn$ and is a different color from the other sequences, giving us r color-focused 2-APs, as desired. ■

We can write $W(m, k)$ for the least n such that any k -coloring of $[n]$ has a monochromatic m -AP. Our proof shows

$$W(3, k) \leq \underbrace{k^{k^{\dots k^{4k}}}}_{k-1}.$$

Gowers showed

$$W(m, 2) \leq 2^{2^{2^{2^{2^{2^m}}}}}.$$

Theorem 2.11 (Wan der Waerden). *For any m and k , there is an n such that any k -coloring of $[n]$ contains a monochromatic m -AP.*

Proof. We induct on m , noting that this is trivial for $m = 1$.

We claim that, for $r \leq k$, there is an $n(m, r)$ such that any k -coloring of $[n]$ has a monochromatic m -AP or r color-focused $(m-1)$ -APs. We prove this by induction on r .

When $r = 1$, take $n = W(m-1, k)$, which exists by the outer inductive hypothesis. Writing $n = n(m, r-1)$, take $n(m, r) = 2nW(m-1, k^{2n})$. A k -coloring of $[n(m, r)]$ has $W(m-1, k^{2n})$ blocks of size $2n$, so there are $B_s, B_{s+t}, \dots, B_{s+(m-1)t}$ that are colored identically, and $r-1$ color-focused $(m-1)$ -APs in B_s , say $\{a_i, a_i + d_i, \dots, a_i + (m-2)d_i\}$ focused at $f = a_i + (m-1)d_i \in B_s$. Now, $\{a_i, a_i + d_i + 2nt, \dots, a_i + (m-2)(d_i + 2nt)\}$ is monochromatic and focused at $f + (m-1)2nt$. Also $\{f, f + 2nt, \dots, f + (m-2)2nt\}$ is monochromatic and focused at $f + (m-1)2nt$. If the foci are the same color as any of the sequences, we have an m -AP. Otherwise, we have r color-focused $(m-1)$ -APs. Thus, the claim holds by induction.

Taking $r = k$, the theorem is true. ■

We alternatively have an infinite version of the theorem:

Theorem 2.12. *Any k -coloring of $\mathbb{N}$ contains an arbitrarily long monochromatic arithmetic progression.*

Another pattern that may arise is a combinatorial line.

Definition 2.13. A *combinatorial line* $L \subseteq [m]^n$ is a subset of the form

$$L = \{(x_1, \dots, x_n) \in [m]^n : x_i = x_j \ \forall i, j \in I, \ x_i = a_i \ \forall i \in [n] \setminus I\}$$

for some $\emptyset \neq I \subseteq [n]$ and fixed constants a_i .

We call the subset I the active coordinates of L . Note that L has a distinguished “last element” given by taken $x_i = m$ for all $i \in I$. We denote this by L^+ . Similarly, we denote the “first element” (where $x_i = 1$ for $i \in I$) by L^- . We’ll say combinatorial lines $L_1, \dots, L_k$ are focused at f if $L_1^+ = \dots = L_k^+ = f$. Given some coloring of $[m]^n$, we’ll say they’re color focused at f if they are focused at f and the $L_i \setminus L_i^+$ are monochromatic of pairwise distinct colors.

Theorem 2.14 (Hales-Jewett). *For any m and k , there is an n such that any k -coloring of $[m]^n$ contains a monochromatic combinatorial line.*

Note that this implies the Van der Waerden theorem: given n and k , we can transform a k -coloring c of $[mn]$ into a k -coloring of $[m]^n$ by $(x_1, \dots, x_n) \mapsto c(x_1 + \dots + x_n)$. For sufficiently large n , this contains a monochromatic combinatorial line, which translates into a monochromatic m -AP in the coloring of $[mn]$. We’ll write $HJ(m, k)$ for the number n given by the theorem.

Proof. We induct on m . When $m = 1$, this is trivial. Suppose that it holds for $m - 1$.

We claim that, for $r \leq k$, there is an n such that any k -coloring of $[m]^n$ contains a monochromatic combinatorial line or r color-focused combinatorial lines. We prove this by induction on r .

When $r = 1$, let $n = HJ(m - 1, k)$ by the outer inductive hypothesis (we’ll get a combinatorial line that is monochromatic when the active coordinates are $1, \dots, m - 1$, meaning that it is color-focused). Suppose that m works for $r - 1$, and consider $n + HJ(m - 1, k^{m^n})$. Write $n' = HJ(m - 1, k^{m^n})$. Given a k -coloring c of $[m]^{n+n'} = [m]^n \times [m]^{n'}$, there are k^{m^n} possible colorings of $[m]^{n'}$. Color $y \in [m]^{n'}$ by the coloring of $[m]^{n'}$ given by $x \mapsto c(x, y)$. By the definition of n' , there is a combinatorial line $L \subseteq [m]^{n'}$ with active coordinates $\emptyset \neq I \subseteq [n']$ such that, for all $a \in [m]^n$ and $b, b' \in L \setminus \{L^+\}$, $c(a, b) = c(a, b')$. Write $c'(a)$ for this color. By the definition of n , there are $r - 1$ color-focused (with respect to c') combinatorial lines $L_1, \dots, L_{r-1}$ with active coordinates $I_1, \dots, I_{r-1}$ and focus f . If f has the same color (under c') as some $L_k \setminus \{L_k^+\}$, then L_k is monochromatic (under c'), so $L_k \times \{b\}$ is monochromatic for some $b \in L \setminus \{L^+\}$. Otherwise, let L'_i be the line passing through (L_i^-, L^-) with active coordinates $I_i \cup I$. Then $L'_1, \dots, L'_{r-1}$ are color-focused at (f, L^+) . Additionally, the line passing through (f, L^+) with active coordinates I is focused at (f, L^+) and monochromatic with a new color except at the focus, so we have r color-focused lines.

The theorem follows by taking $r = k$. ■

2.4 Ultrafilters

An idea from our proof of Ramsey’s theorem is to have some collection $\mathcal{U}$ of subsets of $\mathbb{N}$ that guide our choices. We would want:

- If $A, B \in \mathcal{U}$, $A \cap B \in \mathcal{U}$
- If $A \in \mathcal{U}$ and $A \subseteq B$, then $B \in \mathcal{U}$
- If $A \in \mathcal{U}$ and $A = A_1 \sqcup A_2 \sqcup \dots \sqcup A_k$, then exactly one A_j is in $\mathcal{U}$

Such a collection is called an ultrafilter on $\mathbb{N}$. Note that ultrafilters can never contain $\emptyset$, as $\emptyset = \emptyset \sqcup \emptyset$, which contradicts the last condition. For each $n \in \mathbb{N}$, we have the ultrafilter $\mathcal{U}_n = \{A \subseteq \mathbb{N} : n \in A\}$. The existence of ultrafilters containing only infinite sets is independent from ZF, but is implied by choice (as we will see).

Definition 2.15. A *filter* on $\mathbb{N}$ is a collection $\mathcal{F}$ of subsets of $\mathbb{N}$ satisfying

- $\emptyset \notin \mathcal{F}$
- If $A, B \in \mathcal{F}$, $A \cap B \in \mathcal{F}$
- If $A \in \mathcal{F}$ and $A \subseteq B$, then $B \in \mathcal{F}$

Example. For $A \subseteq \mathbb{N}$, $\{B \subseteq \mathbb{N} : A \subseteq B\}$ is a filter.

Example. The set $\{B \subseteq \mathbb{N} : B \text{ cofinite}\}$ is a filter.

An equivalent definition to our partition condition is the following.

Definition 2.16. An *ultrafilter* is an inclusion-maximal filter.

Zorn’s Lemma implies that any filter can be extended to an ultrafilter. In particular, extending the filter of cofinite subsets of $\mathbb{N}$ to an ultrafilter gives an ultrafilter that contains only infinite subsets (if it contained a finite subset, it would also contain its complement, and hence their intersection, which is $\emptyset$). We see that this is equivalent to our partition condition from the following lemma:

Lemma 2.17. *Let $\mathcal{F}$ be a filter on $\mathbb{N}$. Then $\mathcal{F}$ is an ultrafilter if and only if $\mathcal{F}$ contains A or A^c for all $A \subseteq \mathbb{N}$.*

Proof. Suppose $\mathcal{F}$ contains A or A^c for every $A \subseteq \mathbb{N}$ and is not inclusion-maximal, say $\mathcal{F} \subsetneq \mathcal{F}'$. Let $B \in \mathcal{F}' \setminus \mathcal{F}$. Then, $B^c \in \mathcal{F}$, and so $B^c \in \mathcal{F}'$. Thus, $\emptyset = B \cap B^c \in \mathcal{F}'$, meaning $\mathcal{F}'$ cannot be a filter. Thus, $\mathcal{F}$ is inclusion-maximal.

Conversely, suppose that $\mathcal{F}$ is an inclusion-maximal filter and that $A, A^c \notin \mathcal{F}$. There must be some $B \in \mathcal{F}$ with $B \cap A = \emptyset$: otherwise, $\{C \subseteq \mathbb{N} : C \supseteq B \cap A \text{ for some } B \in \mathcal{F}\}$ is a filter strictly containing $\mathcal{F}$. Similarly, there is some $C \in \mathcal{F}$ with $C \cap A^c = \emptyset$. Now, $B \subseteq A^c$ and $C \subseteq A$, so $\emptyset = B \cap C \in \mathcal{F}$, contradicting the fact that $\mathcal{F}$ is a filter. Thus, $\mathcal{F}$ must contain A or A^c . ■

We'll write $\beta\mathbb{N}$ for the set of all ultrafilters on $\mathbb{N}$. We can think of $\mathcal{U} \in \beta\mathbb{N}$ as an indicator function $2^{\mathbb{N}} \rightarrow \{0, 1\}$, and we can therefore put the product topology on it. Equivalently, we declare $C_A = \{\mathcal{U} \in \beta\mathbb{N} : A \in \mathcal{U}\}$ as the base open (or base closed) sets.

Lemma 2.18. *$\beta\mathbb{N}$ is compact.*

Proof. Suppose we have a collection $\{C_{A_j} : j \in J\}$ of basic closed sets such that every finite intersection is nonempty. Then every finite intersection of the A_j is contained in some (ultra)filter, and so must be nonempty. Let $\mathcal{F} = \{A \subseteq \mathbb{N} : A \supseteq \bigcap_{i \in I} A_i \text{ for some finite } I\}$. Then $\mathcal{F}$ is a filter, and so contained in some ultrafilter $\mathcal{U}$. This $\mathcal{U}$ lies in $\bigcap_{j \in J} C_{A_j}$. ■

Definition 2.19. Given some property $p(x)$ of the natural numbers and an ultrafilter $\mathcal{U}$, we say p holds for $\mathcal{U}$ -most x if $\{x : p(x)\} \in \mathcal{U}$. We write $\forall_{\mathcal{U}} x p(x)$.

Because ultrafilters always contain a set or its complement, exactly one of $\forall_{\mathcal{U}} x p(x)$ and $\forall_{\mathcal{U}} x \neg p(x)$ must hold. Similarly, because ultrafilters are upwards closed, if $\forall_{\mathcal{U}} x (p(x) \wedge q(x))$ holds, then $\forall_{\mathcal{U}} x p(x)$ and $\forall_{\mathcal{U}} x q(x)$ holds. Conversely, if $\forall_{\mathcal{U}} x p(x)$ and $\forall_{\mathcal{U}} x q(x)$ holds, then $\forall_{\mathcal{U}} x (p(x) \wedge q(x))$ holds, because ultrafilters are closed under intersections. Combining these with De Morgan's laws also gives that $\forall_{\mathcal{U}} x (p(x) \vee q(x))$ if and only if $\forall_{\mathcal{U}} x p(x)$ or $\forall_{\mathcal{U}} x q(x)$. In the case that $\mathcal{U} = \mathcal{U}_a$ is principal, $\forall_{\mathcal{U}} x p(x)$ is equivalent to $p(a)$ holding.

Definition 2.20. For ultrafilters $\mathcal{U}$ and $\mathcal{V}$, we define their sum to be

$$\mathcal{U} + \mathcal{V} = \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V}} y x + y \in A\}.$$

In other words,

$$\mathcal{U} + \mathcal{V} = \{A \subseteq \mathbb{N} : \{x : \{y : x + y \in A\} \in \mathcal{V}\} \in \mathcal{U}\}.$$

Notice that this addition need not be commutative. In the case that $\mathcal{U} = \mathcal{U}_n$ and $\mathcal{V} = \mathcal{U}_m$, we have that

$$\forall_{\mathcal{U}} x \forall_{\mathcal{V}} y x + y \in A \iff \forall_{\mathcal{U}} x x + m \in A \iff n + m \in A,$$

so $\mathcal{U}_n + \mathcal{U}_m = \mathcal{U}_{n+m}$. Thus, this definition extends the usual addition of $\mathbb{N}$.

Proposition 2.21. *If $\mathcal{U}$ and $\mathcal{V}$ are ultrafilters, then so is $\mathcal{U} + \mathcal{V}$.*

Proof. Suppose that $A \in \mathcal{U} + \mathcal{V}$ and $A \subseteq B$. For a fixed x , $\{y : x + y \in A\} \subseteq \{y : x + y \in B\}$, so if the former is contained in $\mathcal{V}$, the latter is as well. Thus, $\{x : \{y : x + y \in A\} \in \mathcal{V}\} \subseteq \{x : \{y : x + y \in B\} \in \mathcal{V}\}$. Since the former set is contained in $\mathcal{U}$, the latter set is also contained in $\mathcal{U}$, so $B \in \mathcal{U} + \mathcal{V}$.

Now, suppose that $A, B \in \mathcal{U} + \mathcal{V}$. Once again, for a fixed x , $\{y : x + y \in A \cap B\} = \{y : x + y \in A\} \cap \{y : x + y \in B\}$, so the former is contained in $\mathcal{V}$ if and only if the latter two sets are. Now,

$$\begin{aligned} \{x : \{y : x + y \in A \cap B\} \in \mathcal{V}\} &= \{x : \{y : x + y \in A\} \in \mathcal{V}, \{y : x + y \in B\} \in \mathcal{V}\} \\ &= \{x : \{y : x + y \in A\} \in \mathcal{V}\} \cap \{x : \{y : x + y \in B\} \in \mathcal{V}\} \end{aligned}$$

is contained in $\mathcal{U}$, so $A \cap B \in \mathcal{U} + \mathcal{V}$.

Finally, suppose that $A \notin \mathcal{U} + \mathcal{V}$, so that $\{x : \{y : x + y \in A\} \in \mathcal{V}\} \notin \mathcal{U}$. Then $\mathcal{U}$ must contain

$$\{x : \{y : x + y \in A\} \in \mathcal{V}\}^c = \{x : \{y : x + y \in A\} \notin \mathcal{V}\}.$$

Since $\mathcal{V}$ is an ultrafilter, this set is

$$\{x : \{y : x + y \in A\}^c \in \mathcal{V}\} = \{x : \{y : x + y \notin A\} \in \mathcal{V}\} = \{x : \{y : x + y \in A^c\} \in \mathcal{V}\}.$$

Thus, A^c lies in $\mathcal{U} + \mathcal{V}$. ■

Addition is also, as we would expect, associative. To see this, notice that

$$\begin{aligned}
\mathcal{U} + (\mathcal{V} + \mathcal{W}) &= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V} + \mathcal{W}} y \ x + y \in A\} \\
&= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \ \{t : x + t \in A\} \in \mathcal{V} + \mathcal{W}\} \\
&= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V}} y \forall_{\mathcal{W}} z \ y + z \in \{t : x + t \in A\}\} \\
&= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V}} y \forall_{\mathcal{W}} z \ x + y + z \in A\}
\end{aligned}$$

and that

$$\begin{aligned}
(\mathcal{U} + \mathcal{V}) + \mathcal{W} &= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U} + \mathcal{V}} t \forall_{\mathcal{W}} z \ t + z \in A\} \\
&= \{A \subseteq \mathbb{N} : \{t : \forall_{\mathcal{W}} z \ t + z \in A\} \in \mathcal{U} + \mathcal{V}\} \\
&= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V}} y \ x + y \in \{t : \forall_{\mathcal{W}} z \ t + z \in A\}\} \\
&= \{A \subseteq \mathbb{N} : \forall_{\mathcal{U}} x \forall_{\mathcal{V}} y \forall_{\mathcal{W}} z \ x + y + z \in A\}.
\end{aligned}$$

A less obvious result is the following.

Proposition 2.22. *Let $\mathcal{V}$ be a fixed ultrafilter. Then the map $\mathcal{U} \mapsto \mathcal{U} + \mathcal{V}$ is continuous.*

Proof. For a fixed $A \subseteq \mathbb{N}$, we show that the preimage of the basic open set C_A is open. We have that $\mathcal{U} + \mathcal{V} \in C_A$ if and only if $A \in \mathcal{U} + \mathcal{V}$ if and only if $\{x : \forall_{\mathcal{V}} y \ x + y \in A\} \in \mathcal{U}$. But this set only depends on A and $\mathcal{V}$, which are both fixed. Thus, if $B = \{x : \forall_{\mathcal{V}} y \ x + y \in A\}$, then this occurs if and only if $\mathcal{U} \in C_B$, which is an open set. Thus, the map is continuous. ■

These facts allow us to make use of a powerful topological result.

Lemma 2.23 (Idempotent Lemma). *Let X be a nonempty compact Hausdorff space, and suppose that $+: X \times X \rightarrow X$ is associative and continuous in the left entry. Then there is $x \in X$ such that $x + x = x$.*

Proof. Let S be the collection of nonempty compact subsets $Y \subseteq X$ such that $Y + Y \subseteq Y$ (note S is nonempty, as $X \in S$). We claim that S contains an (inclusion-) minimal element. By Zorn's Lemma, it suffices to show that chains have a lower bound. Given a chain $\{Y_i\}_{i \in I}$ of elements of S , let $Y = \bigcap_{i \in I} Y_i$. Because X is Hausdorff, the Y_i are closed. Since finite intersections of the Y_i are nonempty, the intersection $\bigcap_{i \in I} Y_i$ is nonempty (since X is compact). Additionally, the intersection is closed, and so compact because X is compact. Finally, if $y_1, y_2 \in Y$, then $y_1, y_2 \in Y_i$ for each $i \in I$, so $y_1 + y_2 \in Y_i$ for each i , meaning $y_1 + y_2 \in Y$. Thus, $Y + Y \subseteq Y$, so $Y \in S$, and we have a lower bound for the chain.

Now, let Y be a minimal element of S , and take $x \in Y$. Because Y is compact and $+$ is left-continuous, $Y + x$ is nonempty and compact. Additionally, $Y + x \subseteq Y$, so

$$(Y + x) + (Y + x) \subseteq Y + (Y + x) = (Y + Y) + x \subseteq Y + x,$$

so $Y + x \in S$. Since $Y + x \subseteq Y + Y \subseteq Y$, we must have that $Y + x = Y$. Thus, if $Z = \{y \in Y : y + x = x\}$, then Z is nonempty. Since Z is the preimage of the closed set $\{x\}$ under the continuous map $\cdot + x$, it is closed, and thus compact. If $y_1, y_2 \in Z$, then $(y_1 + y_2) + x = y_1 + (y_2 + x) = y_1 + x = x$, so $y_1 + y_2 \in Z$. Thus, $Z \in S$. Since $Z \subseteq Y$ and Y is inclusion-minimal, $Z = Y$. In particular, $x \in Z$, so $x + x = x$, as desired. ■

Definition 2.24. If $X \subseteq \mathbb{N}$, let

$$\text{FS}(X) = \left\{ \sum_{i=1}^n x_i : x_i \in X, n \geq 1, x_i \text{ pairwise distinct} \right\}$$

denote the set all values that are finite sums of elements of X .

In the proof of the following theorem, we will be able to use ultrafilters as a way to “pick out” what to do.

Theorem 2.25 (Hindman). *For any k -coloring of $\mathbb{N}$, there is an infinite $X \subseteq \mathbb{N}$ such that $\text{FS}(X)$ is monochromatic.*

Proof. Let $\mathcal{U} \in \beta\mathbb{N}$ with $\mathcal{U} + \mathcal{U} = \mathcal{U}$. Note that $\mathcal{U}$ is nonprincipal (as $\mathcal{U}_n + \mathcal{U}_n = \mathcal{U}_{2n}$, so if $\mathbb{N}$ doesn't include 0, then $\mathcal{U} \neq \mathcal{U}_n$ for all n). In particular, $\mathcal{U}$ does not contain any singletons. Now, for $a \in A \in \mathcal{U}$, $A = \{a\} \sqcup (A \setminus \{a\})$, so we must have that $A \setminus \{a\} \in \mathcal{U}$, as $\mathcal{U}$ is an ultrafilter.

For $B \in \mathcal{U}$, let $B' = \{n \in \mathbb{N} : B - n \in \mathcal{U}\}$ (where we throw out nonpositive elements from B' if necessary). Because $B \in \mathcal{U} = \mathcal{U} + \mathcal{U}$, we must have that $\{x : \{y : x + y \in B\} \in \mathcal{U}\} \in \mathcal{U}$ by the definition of ultrafilter addition. However, $\{y : x + y \in B\} = B - x$, so this is the set $\{x : B - x \in \mathcal{U}\} = B'$. Thus, $B' \in \mathcal{U}$.

Suppose we have a k -coloring of $\mathbb{N}$ with color classes $C_1, \dots, C_k$. One of these must be in $\mathcal{U}$, say $C_j = A_0$. Let $a_1 \in A_0 \cap A'_0$, and let $A_1 = A_0 \cap (A_0 - a_1) \setminus \{a_1\} \in \mathcal{U}$. We have that $A_1 \subsetneq A_0$, $a_1 + A_1 \subseteq A_0$, and $A_1 \in \mathcal{U}$. Repeat this to get $A_0 \supsetneq A_1 \supsetneq A_2 \supsetneq \dots$ all contained in $\mathcal{U}$ and $a_i \in A_{i-1}$ with $a_i + A_i \subseteq A_{i-1}$. Let $X = \{a_1, a_2, \dots\} \subseteq A_0$. By construction, $\text{FS}(X) \subseteq A_0$. Since $A_0 = C_j$ is a color class of our coloring, $\text{FS}(X) \subseteq C_j$ is monochromatic. ■

2.5 Discrete Dynamical Systems

Definition 2.26. Let $A \subseteq \mathbb{N}$. We say A is *syndetic* if there is some $m \in \mathbb{N}$ such that $A \cap [k, k + m]$ is nonempty for all $k \in \mathbb{N}$ (i.e. if A has bounded gaps). We say A is *thick* if it intersects every syndetic set (i.e. if A contains arbitrarily long intervals). We say A is *piecewise syndetic* if it is the intersection of a syndetic set and a thick set (i.e. if A is syndetic on some given arbitrarily long intervals).

We may hope that any coloring of $\mathbb{N}$ contains a monochromatic syndetic set. However, this is quite easy to avoid. For example, color 1 red, 2 blue, 3 – 4 red, 5 – 6 blue, 7 – 9 red, 10 – 12 blue, and so on. Because of this, we look at piecewise syndetic sets instead.

Definition 2.27. A *discrete dynamical system* is a compact metric space X together with a continuous map $T : X \rightarrow X$.

Definition 2.28. Let (X, T) be a discrete dynamical system. We say $x \in X$ is *recurrent* if the set $\{n \in \mathbb{N} : T^n(x) \in V\}$ is nonempty for any open $V \ni x$ (where we take $\mathbb{N}$ to exclude 0). We say x is *uniformly recurrent* if the set $\{n \in \mathbb{N} : T^n(x) \in V\}$ is syndetic for any open $V \ni x$.

Definition 2.29. If (X, T) is a discrete dynamical system, a *subsystem* of X is a nonempty closed subset $Y \subseteq X$ with $T(Y) \subseteq Y$.

Theorem 2.30. Every discrete dynamical system (X, T) contains a minimal subsystem, and every point in a minimal subsystem is recurrent. Moreover, every point in a minimal subsystem is uniformly recurrent.

Proof. The existence of minimal subsystems follows from Zorn's Lemma (the intersection of a chain of subsystems is itself a subsystem because it must be nonempty, as finite intersections from the chain are nonempty and X is compact). Thus, assume WLOG that X is minimal.

Take $x \in X$, and let $Y_k = \overline{\{T^k x, T^{k+1} x, T^{k+2} x, \dots\}} \subseteq X$. We claim that Y_k is a subsystem. If $z \in Y$ is $T^n x$ for some $n \geq k$, then $Tz = T^{n+1} x \in Y$. Otherwise, z is a limit point $\lim_{i \rightarrow \infty} T^{n_i} x$ for $n_1 < n_2 < \dots$. But T is continuous, so $Tz = T \lim_{i \rightarrow \infty} T^{n_i} x = \lim_{i \rightarrow \infty} T^{n_i+1} x \in Y$. Now, since X is minimal, $X = Y_k$. In particular, if $V \ni x$ is open and $T^n x \notin V$ for $n \geq 1$, then $Y_1 \subseteq X \setminus V \subsetneq X$, a contradiction. Thus, x must be recurrent.

To see that x is uniformly recurrent, let $V \subseteq X$ be open, and note that $X = \bigcup_{n \in \mathbb{N}} T^{-n}(V)$ (if there was some z such that $T^n z$ avoids V , then $\overline{\{T^n(z) : n \in \mathbb{N}\}} \subseteq X \setminus V$ would be a strictly smaller subsystem). Since X is compact, we can take a finite subcover $X = \bigcup_{n \in F} T^{-n}(V)$. Now, for any $m \in \mathbb{N}$, there is a $k \in F$ such that $T^m(x) \in T^{-k}(V)$, meaning that $k + m \in \{n : T^n x \in V\}$. Because F is finite, this means the set $\{n : T^n x \in V\}$ is syndetic (it must always intersect intervals of length $\geq \max F + 1$). ■

Theorem 2.31. For any k -coloring of $\mathbb{N}$, one of the color classes is piecewise syndetic.

Proof. Let $X = [k]^\mathbb{N}$ be the space of k -colorings of $\mathbb{N}$ with the metric

$$d(c, c') = \frac{1}{\min\{n : c(n) \neq c'(n)\}}$$

(with $d(c, c') = 0$ when c and c' are equal). With this metric, lying in a $1/k$ -ball around a point means agreeing in the first k coordinates. This induces the product topology on X , and so is compact. Define $T : X \rightarrow X$ by $T(c) : n \mapsto c(n + 1)$, which “shifts a coloring to the left”. Note that $d(T(c), T(c')) \leq 2d(c, c')$, so T is continuous.

Since lying in a $1/k$ -ball means agreeing in the first k coordinates, a point $c \in X$ is syndetic iff every string in c occurs along a syndetic set (the reason this is every string, rather than the strings starting from the beginning, is that we can apply T enough times to bring any string to the front, and we must still have a uniformly recurrent point). Fix a coloring $f \in X$, take $Y = \{T^n(f) : n \in \mathbb{N}\}$ a subsystem of X , and let $h \in Y$ be uniformly recurrent. Let $j \in [k]$ occur syndetically in h (i.e. the substring “ j ” appears along a syndetic set), say with bounded gaps of length ℓ .

For any $\varepsilon > 0$, there is some $m \in \mathbb{N}$ with $d(T^m(f), h) < \varepsilon$ (since $h \in Y$). Taking $\varepsilon = 1/n$, this means that $T^m(f)(i) = h(i)$ for $1 \leq i \leq n$. Since j occurs syndetically in $\{h(1), \dots, h(n)\}$, it occurs syndetically in $\{T^m(f)(1), \dots, T^m(f)(n)\} = \{f(m+1), \dots, f(m+n)\}$. Since we can take n to be arbitrarily large, j is piecewise syndetic in f , as desired. $\blacksquare$

3 Discrete Fourier Analysis

3.1 Definitions

Our object of study will be functions $f : \mathbb{Z}/n \rightarrow \mathbb{C}$. We'll denote the space of all such functions by $F_n(\mathbb{C})$. For $f \in F_n(\mathbb{C})$, we write $\mathbb{E}_x f(x) = \frac{1}{n} \sum_{x \in \mathbb{Z}/n} f(x)$. If $\omega = e^{2\pi i/n}$, we recall that

$$\sum_{r=0}^{n-1} \omega^{rx} = \begin{cases} n & x = 0 \\ 0 & \text{else} \end{cases},$$

where x is taken modulo n . We'll write $\delta_x = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$.

Definition 3.1. Let $f, g \in F_n(\mathbb{C})$ and write $\omega = e^{2\pi i/n}$.

- The *Fourier Transform* of f is a function $\mathbb{Z}/n \rightarrow \mathbb{C}$ given by $\hat{f}(r) = \mathbb{E}_x f(x) \omega^{-rx}$. We denote the space of all Fourier transforms by $\hat{F}_n(\mathbb{C})$.
- The *inner product* of f and g is $\langle f, g \rangle = \mathbb{E}_x f(x) \overline{g(x)}$.
- The ℓ^p *norm* of f is $\|f\|_p = (\mathbb{E}_x |f(x)|^p)^{1/p}$.
- The *convolution* of f and g is an element of $F_n(\mathbb{C})$ given by $(f * g)(x) = \mathbb{E}_{y+z=x} f(y)g(z)$. $f * g$ lies in $F_n(\mathbb{C})$.

There is no actual difference between $F_n(\mathbb{C})$ and $\hat{F}_n(\mathbb{C})$ (we will see that every function is a Fourier transform of some function). However, we keep track of which space a function lies in, and use the following definitions for elements of $\hat{F}_n(\mathbb{C})$:

Definition 3.2. Let $f, g \in \hat{F}_n(\mathbb{C})$ and write $\omega = e^{2\pi i/n}$.

- The *Fourier Transform* of f is an element of $F_n(\mathbb{C})$ given by $\hat{f}(r) = \sum_x f(x) \omega^{rx}$.
- The *inner product* of f and g is $\langle f, g \rangle = \sum_x f(x) \overline{g(x)}$.
- The ℓ^p *norm* of f is $\|f\|_p = (\sum_x |f(x)|^p)^{1/p}$.
- The *convolution* of f and g is an element of $\hat{F}_n(\mathbb{C})$ given by $(f * g)(x) = \sum_{y+z=x} f(y)g(z)$.

The reason for these alternate definitions is that we now have the following properties:

Proposition 3.3. Let $f, g \in F_n(\mathbb{C})$. Then,

- (a) $f = \widehat{\hat{f}}$
- (b) $\langle f, g \rangle = \langle \hat{f}, \hat{g} \rangle$
- (c) $\|f\|_2 = \|\hat{f}\|_2$
- (d) $\widehat{f * g} = \hat{f} \cdot \hat{g}$

Proof.

- (a) We have that

$$\hat{f}(x) = \sum_r \hat{f}(r) \omega^{rx} = \sum_r \left(\mathbb{E}_y f(y) \omega^{-ry} \right) \omega^{rx} = \mathbb{E}_y f(y) \sum_r \omega^{r(x-y)} = n \mathbb{E}_y f(y) \delta_{x-y} = f(x).$$

(b) We have that

$$\begin{aligned}\langle \hat{f}, \hat{g} \rangle &= \sum_x \hat{f}(x) \overline{\hat{g}(x)} = \sum_x \left(\mathbb{E}_y f(y) \omega^{-xy} \right) \left(\mathbb{E}_z \overline{g(z)} \omega^{xz} \right) = \frac{1}{n^2} \sum_{x,y,z} f(y) \overline{g(z)} \omega^{x(z-y)} \\ &= \frac{1}{n^2} \sum_{y,z} f(y) \overline{g(z)} \left(\sum_x \omega^{x(z-y)} \right) = \frac{1}{n^2} \sum_{y,z} f(y) \overline{g(z)} (n \delta_{z-y}) = \frac{1}{n} \sum_y f(y) \overline{g(y)} = \langle f, g \rangle.\end{aligned}$$

(c) Take $f = g$ above.

(d) We have that

$$\begin{aligned}(\widehat{f * g})(r) &= \mathbb{E}_x (f * g)(x) \omega^{-rx} = \mathbb{E}_x \left(\mathbb{E}_{y+z=x} f(y) g(z) \right) \omega^{-rx} = \frac{1}{n^2} \sum_x \sum_{y+z=x} f(y) g(z) \omega^{-rx} \\ &= \frac{1}{n^2} \sum_{y,z} f(y) g(z) \omega^{-r(y+z)} = \frac{1}{n^2} \left(\sum_y f(y) \omega^{-ry} \right) \left(\sum_z g(z) \omega^{-rz} \right) = \hat{f}(r) \hat{g}(r).\end{aligned}$$

■

The idea of a Fourier transform is that the Fourier transform of a random function should be very close to 0, so we can use nonzero values of the Fourier transform to pick out structure. In particular, for $A \subseteq \mathbb{Z}/n$, we'll be interested in the indicator function $A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$. We write $\delta(A) = |A|/n$ for the density of A . Note that

$$\hat{A}(0) = \mathbb{E}_x A(x) \omega^0 = \delta(A)$$

and that

$$\sum_r |\hat{A}(r)|^2 = \|\hat{A}\|_2^2 = \|A\|_2^2 = \mathbb{E}_x |A(x)|^2 = \delta(A).$$

3.2 Roth's Theorem

Theorem 3.4 (Szemerédi). *Let $\delta > 0$ and $m \in \mathbb{N}$. Then there is $n \in \mathbb{N}$ such that every $A \subseteq [n]$ with $|A| \geq \delta n$ contains an m -AP.*

We will focus on the case $m = 3$:

Theorem 3.5 (Roth). *Let $\delta > 0$. Then there is $n \in \mathbb{N}$ such that every $A \subseteq [n]$ with $|A| \geq \delta n$ contains a 3-AP.*

We will make use of the following key proposition.

Proposition 3.6. *Fix $\delta > 0$, N large, and $A \subseteq [N]$ with $|A| \geq \delta N$. Then, A has a 3-AP or there is a subprogression $P \subseteq [N]$ of length N' (depending on N and δ) such that $|A \cap P| \geq (\delta + c(\delta))|P|$, where $N' \rightarrow \infty$ as $N \rightarrow \infty$ and $c(\delta)$ is bounded away from 0 whenever δ is bounded away from 0 (that is, for all $\varepsilon > 0$, there is $\varepsilon' > 0$ such that $c(\delta) > \varepsilon'$ whenever $\delta > \varepsilon$).*

This proposition essentially says that, for sufficiently large N , we can either find a 3-AP in A , or pass to a set of higher density within another large AP (which helps because APs within APs are still in AP).

Proof of Roth's Theorem. Let $\delta_* = \inf\{\delta > 0 : \text{Roth's Theorem holds}\}$, and suppose for the sake of contradiction that $\delta_* > 0$.

Suppose first that Roth's Theorem does not hold for δ_* . Then, for all N , we can find a subset $A \subseteq [N]$ of density δ_* containing no 3-APs. But then the proposition means that we can find a subset $A' \subseteq [N']$ of density $\delta_* + c(\delta_*)$, which must necessarily also not contain any 3-APs. However, since $N' \rightarrow \infty$ as $N \rightarrow \infty$, this means that Roth's Theorem does not hold for $\delta_* + c(\delta_*) > \delta_*$, contradicting the minimality of δ_* .

Thus, Roth's Theorem holds for δ_* . Take ε' so that $c(\delta) > \varepsilon'$ for $\delta > \delta_*/2$ (which is possible because c is bounded away from 0). If $\varepsilon < \min(\varepsilon', \delta_*/2)$, then $\delta_* - \varepsilon > \delta_*/2$, so $c(\delta_* - \varepsilon) > \varepsilon' > \varepsilon$. Now, for sufficiently large N , if there is some $A \subseteq [N]$ of density $\geq \delta_* - \varepsilon$ with no 3-AP, we can use the proposition to find an AP $P \subseteq [N]$ of length N' and $|A \cap P| \geq (\delta_* - \varepsilon + c(\delta_* - \varepsilon))|P| > \delta_*|P|$, so we can find a 3-AP after taking N large enough, since Roth's Theorem holds for δ_* . But $\delta_* - \varepsilon < \delta_*$, a contradiction. Thus, we must have that $\delta_* = 0$. ■

Finding the subprogression P is where we will take advantage of Fourier Analysis. We'll assume that n is odd so that $z \mapsto z/2$ is a well-defined bijection, and write $A_2(z) = A(z/2)$.

We want 3-APs within A , which is the same as looking for $x, y, z \in A$ with $x + y = 2z$. Thus, we are interested in

$$\mathbb{E}_{x+y=2z} A(x)A(y)A(z),$$

as a nonzero value means that A contains a 3-AP. Now,

$$\begin{aligned} \mathbb{E}_{x+y=2z} A(x)A(y)A(z) &= \mathbb{E}_{x+y=z} A(x)A(y)A_2(z) = \mathbb{E}_z (A * A)(z)A_2(z) \\ &= \langle A * A, A_2 \rangle = \langle \widehat{A * A}, \hat{A}_2 \rangle = \langle \hat{A}^2, \hat{A}_2 \rangle \\ &+ \sum_r \hat{A}(r)^2 \overline{\hat{A}_2(r)} = \sum_r \hat{A}(r)^2 \overline{\hat{A}(2r)} \\ &= \sum_r \hat{A}(r)^2 \hat{A}(-2r) = \delta(A)^3 + \sum_{r \neq 0} \hat{A}(r)^2 \hat{A}(-2r). \end{aligned}$$

The $\delta(A)^3$ term is what we would expect randomly, if x, y, z were included independently, and the remaining sum comes from the structure of A . Now,

$$\begin{aligned} \left| \sum_{r \neq 0} \hat{A}(r)^2 \hat{A}(-2r) \right| &\leq \max_{r \neq 0} |\hat{A}(r)| \cdot \sum_{r \neq 0} |\hat{A}(r)| \cdot |\hat{A}(-2r)| \\ &\leq \max_{r \neq 0} |\hat{A}(r)| \cdot \left(\sum_r |\hat{A}(r)|^2 \right)^{1/2} \left(\sum_r |\hat{A}(-2r)|^2 \right)^{1/2} \\ &= \left(\max_{r \neq 0} |\hat{A}(r)| \right) \left(\sum_r |\hat{A}(r)|^2 \right) = \left(\max_{r \neq 0} |\hat{A}(r)| \right) \cdot \delta(A). \end{aligned}$$

Thus,

$$\mathbb{E}_{x+y=2z} A(x)A(y)A(z) \geq \delta(A)^3 - \delta(A) \cdot \max_{r \neq 0} |\hat{A}(r)|.$$

If $|\hat{A}(r)| < \delta(A)^2$ for all $r \neq 0$, we are done. Otherwise, suppose that $|\hat{A}(r)| > \delta(A)^2$ for some $r \neq 0$. We will use this r to identify a subprogression where A has higher density. The way we do this is by finding subprogressions such that ω^{-rx} points “in the same direction” for x in the same subprogression.

Lemma 3.7. Fix r , and let $N \geq 1$ and $\varepsilon > 0$. Then there is N' (depending on N and ε) with $N' \rightarrow \infty$ as $N \rightarrow \infty$ for any fixed ε so that we can find a partition $[N] = \bigcup_{j=1}^J P_j \cup E$, where the P_j are APs of length N' , E has size independent of N , and $\max_{x,y \in P_j} |\omega^{rx} - \omega^{ry}| \in O(\varepsilon)$ for all j .

Proof. The content of this theorem is that $N' \rightarrow \infty$ as $N \rightarrow \infty$, as we could otherwise take $N' = 1$ and be done. In particular, we can assume that N is as large as we like, as we only care about the long-term behavior.

When $r = 1$, divide the unit circle into segments of length ε , and take the P_j s to be the powers of ω lying within each segment. We can make them all have the same length N' by throwing at most one element into E (so that $|E| < 1/\varepsilon$ is independent of N).

For the general case, take h so that ω^{rh} is very close to 1 (which we can do since we only need this to work for sufficiently large N). Then, multiplying the sets from the $r = 1$ case by h gives a solution for this case. $\blacksquare$

If $f(x) = A(x) - \delta(A)$, then $\hat{f}(s) = \begin{cases} 0 & s = 0 \\ \hat{A}(s) & s \neq 0 \end{cases}$. Thus,

$$\delta(A)^2 \leq |\hat{A}(r)| = |\hat{f}(r)| = \left| \mathbb{E}_x (A(x) - \delta(A)) \omega^{-rx} \right| \leq \frac{1}{N} \sum_{j=1}^J \left| \sum_{x \in P_j} (A(x) - \delta(A)) \omega^{-rx} \right| + O(\varepsilon),$$

where the $O(\varepsilon)$ term comes from the error set E (it gives an error in $O(\varepsilon)$ because of how we constructed it). Since the ω^{-rx} point roughly in the same direction for $x \in P_j$ (with an error of $O(\varepsilon)$), we have that

$$\delta(A)^2 \leq \sum_{j=1}^J \left| \sum_{x \in P_j} A(x) - \delta(A) \right| + O(\varepsilon).$$

Thus, $\sum_{j=1}^J \left| \sum_{x \in P_j} A(x) - \delta(A) \right| \geq \delta(A)^2 N + O(\varepsilon)$. Additionally, $\sum x A(x) - \delta(A) = 0$, so we must have $\sum_{j=1}^J \sum_{x \in P_j} A(x) - \delta(A) \in O(\varepsilon)$. Adding these, and using the fact that $x + |x| = 2 \max(x, 0)$,

$$\sum_{j=1}^J 2 \max \left(\sum_{x \in P_j} A(x) - \delta(A), 0 \right) \geq \delta(A)^2 N + O(\varepsilon).$$

For sufficiently small ε , this is $\geq \delta(A)^2 N/2$. Thus, by the pigeonhole principle, there is some j with $\max \left(\sum_{x \in P_j} A(x) - \delta, 0 \right) \geq \delta^2 |P_j|/4$. Thus, $|A \cap P_j| \geq (\delta(A) + \delta(A)^2/4) |P_j|$, and $\delta(A)^2/4$ is bounded away from 0 when $\delta(A)$ is bounded away from 0, so this completes the proof of the proposition.

3.3 Behrend's Construction

Let $S_r = \{(x_1, \dots, x_k) \in [m]^k : \sum x_i^2 = r\}$. Since this is part of the surface of a sphere, it cannot contain 3 colinear points. Additionally, $S_r = \emptyset$ for $r > m^2 k$. Thus, there must be some r with $|S_r| \geq m^k/m^2 k$. We can define $\phi : S_r \rightarrow \mathbb{Z}$ by

$$(x_1, \dots, x_k) \mapsto x_1 + (2m)x_2 + (2m)^2 x_3 + \dots + (2m)^{k-1} x_k \leq m \frac{(2m)^k - 1}{2m - 1} \leq (2m)^k.$$

We claim that $\text{im } \phi$ contains no 3-APs. If $\phi(x)$, $\phi(y)$ and $\phi(z)$ were in AP, we could reduce mod $2m$ to see that x_1, y_1, z_1 are in AP mod $2m$, and so must be in AP (since they lie in $[m]$). Reducing mod $(2m)^2$ shows that $2mx_2, 2my_2, 2mz_2$ are in AP mod $(2m)^2$, so x_2, y_2, z_2 are in AP mod $2m$, and so are in AP. Continuing, we see that x, y, z are colinear, which we know is impossible.

Now, if $k = \sqrt{\log n}$ and $m = \frac{1}{2} e^{\sqrt{\log n}}$ (so that $n = (2m)^k$), we have that

$$\frac{m^k}{m^2 k} = \frac{n}{2\sqrt{\log n}} \cdot \frac{1}{\frac{1}{4} e^{2\sqrt{\log n}} \sqrt{\log n}} \geq n e^{-c\sqrt{\log n}}.$$

This gives a 3-AP-free subset of $[n]$ of density $\geq e^{-c\sqrt{\log n}}$.

4 Polynomial Methods

4.1 Point-Line Incidences

If $\mathcal{P}$ is a set of points in $\mathbb{R}^2$ and $\mathcal{L}$ is a set of lines in $\mathbb{R}^2$, we say that an *incidence* is a pair $(p, \ell) \in \mathcal{P} \times \mathcal{L}$ such that p lies on ℓ . Note that p need not lie at an intersection of lines, and that if p does lie on an intersection of lines, we have an incidence for each line in that intersection. We write $I(\mathcal{P}, \mathcal{L})$ for the number of incidences.

Theorem 4.1 (Szemerédi-Trotter). *If $|\mathcal{P}| = n$ and $|\mathcal{L}| = m$, then*

$$I(\mathcal{P}, \mathcal{L}) \in O(n^{2/3} m^{2/3} + n + m).$$

The n and m terms catch boundary cases when one of n and m is significantly larger than the other (for example, when there are n points and 1 line, or 1 point and m lines). In the general case, $I(\mathcal{P}, \mathcal{L}) \in O(n^{2/3} m^{2/3})$.

In the case that $n = m^\lambda$, Szemerédi-Trotter says that

$$I(\mathcal{P}, \mathcal{L}) \in \begin{cases} O(n) & \lambda < 1/2 \\ O(n^{2/3} m^{2/3}) & 1/2 \leq \lambda \leq 2 \\ O(m) & 2 < \lambda \end{cases}.$$

The power of point-line incidences is that we can transform questions about the size of sets into questions about incidences.

For example, let $A \subseteq \mathbb{R}$, and define $A + A = \{a + b : a, b \in A\}$ and $A \cdot A = \{ab : a, b \in A\}$. In the case that $A = [n]$, $|A + A| = 2n - 1$ is small (in $O(n)$). Meanwhile, if $A = \{2^i : i \in [n]\}$, then $|A \cdot A| = 2n - 1$ is small (in $O(n)$). However, it is not possible that both $A + A$ and $A \cdot A$ are small.

Theorem 4.2 (Elekes). $\max(|A + A|, |A \cdot A|) \in \Omega(|A|^{5/4})$.

Proof. Let $A \subseteq \mathbb{R}$. Define $\mathcal{P} = \{(c, d) : c \in A + A, d \in A \cdot A\}$ and $\mathcal{L} = \{y = a(x - a') : a, a' \in A\}$. We see that $|\mathcal{P}| = |A + A| \cdot |A \cdot A|$ and $|\mathcal{L}| = |A|^2$. Additionally, the line $y = a(x - a')$ contains the points $(b + a', ba)$ for all $b \in A$, so there are $\geq |A| \cdot |\mathcal{L}| = |A|^3$ incidences. Thus,

$$|A|^3 \leq I(\mathcal{P}, \mathcal{L}) \in O(|\mathcal{P}|^{2/3} |\mathcal{L}|^{2/3} + |\mathcal{P}| + |\mathcal{L}|).$$

Now,

$$|\mathcal{L}| = |A|^2 \leq |A + A| \cdot |A \cdot A| \leq |A|^4 = \mathcal{L}^4,$$

so the $|\mathcal{P}|^{2/3} |\mathcal{L}|^{2/3}$ term dominates by our discussion above. Thus, there is some c (not depending on A) so that

$$|A|^3 \leq c \cdot |\mathcal{P}|^{2/3} |\mathcal{L}|^{2/3} = c |A + A|^{2/3} |A \cdot A|^{2/3} |A|^{4/3},$$

meaning that

$$|A|^{5/2} \leq c |A + A| \cdot |A \cdot A|.$$

Thus, one of $|A + A|$ and $|A \cdot A|$ has size in $\Omega(|A|^{5/4})$. ■

How can we prove the Szemerédi-Trotter theorem? Given any graph, we can draw it in the plane (possibly with edge crossings). We can modify this so that no 3 edges intersect except at vertices, and so that edges only intersect the vertices that they are incident to. We say that the crossing number of G is the minimal number of transverse intersections among all such drawings. For example, $\text{cr}(K_5) = 1$ (draw K_4 in a planar way, and insert one other point, we will only need 1 transverse intersection).

If $m > 3n$, we may delete an edge from every transverse crossing to get a planar graph with $m' = m - \text{cr}(G)$ edges. This must satisfy $m - \text{cr}(G) = m' \leq 3n$, so that $\text{cr}(G) \geq m - 3n$.

Now, if $m \geq 4n$, let G' be a random induced subgraph of G where we retain a vertex with probability p . Then,

$$\mathbb{E}(\text{cr}(G')) \geq \mathbb{E}(m' - 3n') = \mathbb{E}(m') - 3\mathbb{E}(n') = p^2 m - 3pn.$$

Additionally, a transverse crossing survives in G' only if each of the 4 vertices from the edges survive, so

$$\mathbb{E}(\text{cr}(G')) \leq p^4 \text{cr}(G).$$

Thus,

$$\text{cr}(G) \geq \frac{m}{p^2} - \frac{3n}{p^3}.$$

Taking $p = 4n/m$, $\text{cr}(G) \geq m^3/64n^2$.

Proof of Szemerédi-Trotter. Suppose that every line is incident to at least 2 points (otherwise, the number of incidences is in $O(m)$). View the point-line configuration as a graph G , with the points being vertices and lines being edges. Delete the two unbounded rays from each line.

For each line, arbitrarily pick some orientation (note that every edge along that line will be oriented the same way), and associate each incidence with the edge coming out of it. This is impossible exactly when the point is the last one on the line, so $I(\mathcal{P}, \mathcal{L}) = \#E(G) + m$. Now, any two lines intersect at most once, so $\text{cr}(G) \leq m^2$. If $I(\mathcal{P}, \mathcal{L}) - m < 4n$, then $I(\mathcal{P}, \mathcal{L}) \in O(m + n)$. Otherwise,

$$m^2 \geq \text{cr}(G) \geq \frac{(I(\mathcal{P}, \mathcal{L}) - m)^3}{64n^2},$$

so

$$I(\mathcal{P}, \mathcal{L}) \leq m + 4n^{2/3} m^{2/3} \in O(n^{2/3} m^{2/3} + n + m). ■$$

This is optimal: consider an $N^2 \times N$ grid of points and lines through each point of slope $1, \dots, N$. There are N^3 points, $O(N^3)$ lines, and each point has N lines, meaning that there are $O(N^4)$ incidences.

4.2 Joints in $\mathbb{R}^3$

Definition 4.3. Let $\mathcal{L}$ be a set of lines in $\mathbb{R}^3$. A joint of $\mathcal{L}$ is a point contained in 3 non-coplanar lines. We write $J(\mathcal{L})$ for the number of joints.

For example, if we have N planes in $\mathbb{R}^3$, there are $O(N^2)$ intersections of two planes (i.e. lines) and $O(N^3)$ intersections of 3 planes (i.e. points).

More generally, for a field $\mathbb{F}$, we define a line in $\mathbb{F}^n$ as a set $\{at + b : t \in \mathbb{F}\}$ for fixed $a, b \in \mathbb{F}^n$ with a nonzero.

Definition 4.4. Let $\mathbb{F}$ be a field. Write $\text{Poly}_D(\mathbb{F}^n)$ for the $\mathbb{F}$ -vector space of $\mathbb{F}$ -polynomials in n variables with degree at most D .

Proposition 4.5.

- (a) If $S \subseteq \mathbb{F}^n$ is finite and $\dim \text{Poly}_D(\mathbb{F}^n) > |S|$, then there is a nonzero $P \in \text{Poly}_D(\mathbb{F}^n)$ that vanishes on S .
- (b) $\dim \text{Poly}_D(\mathbb{F}^n) = \binom{D+n}{n} \geq \frac{D^n}{n!}$.
- (c) If $S \subseteq \mathbb{F}^n$ is finite, there exists a nonzero polynomial that vanishes on S of degree $\leq n|S|^{1/n}$.
- (d) If $P \in \text{Poly}_D(\mathbb{F})$ and $x_1 \in \mathbb{F}$, then there are $P_1 \in \text{Poly}_{D-1}(\mathbb{F})$ and $r \in \mathbb{F}$ with $P = (x - x_1)P_1 + r$.
- (e) If $P \in \text{Poly}_D(\mathbb{F})$ vanishes at $D+1$ points, $P = 0$.
- (f) If $P \in \text{Poly}_D(\mathbb{F}^n)$ vanishes at $D+1$ points on a line, then P vanishes on the whole line.

Proof.

- (a) The evaluation map $\text{Poly}_D(\mathbb{F}^n) \rightarrow \mathbb{F}^S$ must have nontrivial kernel by dimension considerations.
- (b) The set $\{x_1^{D_1} \cdots x_n^{D_n} : \sum D_i \leq D\}$ is a basis of size $\binom{D+n}{n}$.
- (c) This follows from the previous two results.
- (d) This is a standard fact in algebra.
- (e) Note $\dim \text{Poly}_D(\mathbb{F}) = D+1$. Let the points be $x_1, \dots, x_{D+1}$, and define $F_i = \prod_{j \neq i} (x - x_j) / \prod_{j \neq i} (x_i - x_j)$. Then $F_i(x_j) = \delta_{i,j}$, which means that the F_i are linearly independent, and thus spanning. This means the only function vanishing on all of the x_i is 0.
- (f) Apply the previous part to the polynomial $P(at + b)$, where our line is $\{at + b : t \in \mathbb{F}\}$.

■

The last fact means that polynomials are highly restrictive on lines. Thus, when thinking about sets involving lines, it is smart to consider polynomials that vanish on those sets.

Lemma 4.6. If $\mathcal{L}$ is a set of lines in $\mathbb{R}^3$ with J joints, then there is a line with $\leq 3J^{1/3}$ joints.

Proof. Let P be the lowest degree nonzero polynomial vanishing at all joints. We know that $\deg P \leq 3J^{1/3}$. If all lines have $> 3J^{1/3}$ points, then P vanishes along every line. Now, $\nabla P = (\frac{\partial P}{\partial x}, \frac{\partial P}{\partial y}, \frac{\partial P}{\partial z})$ must vanish at all joints (as the directional derivative of P vanishes in 3 linearly independent directions). Thus, each partial derivative vanishes at all joints and has a lower degree than P , meaning that each partial derivative is 0. Thus, P is constant, a clear contradiction. ■

Theorem 4.7. If $\mathcal{L}$ is a set of lines in $\mathbb{R}^3$, $J(\mathcal{L}) \in O(|\mathcal{L}|^{3/2})$.

Proof. Let $J(L)$ denote the maximum number of joints for L lines. We know some line ℓ must contain $\leq 3J(L)^{1/3}$ joints, and the remaining lines have $\leq J(L-1)$ joints between them. Thus, $J(L) \leq J(L-1) + 3J(L)^{1/3}$. Recursing,

$$J(L) \leq 3J(1)^{1/3} + 3J(2)^{1/3} + \cdots + 3J(L)^{1/3} \leq 3LJ(L)^{1/3}.$$

Thus, $J(L) \leq (3L)^{3/2}$. ■

4.3 Nikodym and Kakeya Sets

We will turn towards considering finite fields in particular.

Proposition 4.8. *If $P \in \text{Poly}_{q-1}(\mathbb{F}_q^n)$ vanishes everywhere, then $P = 0$.*

Proof. We induct on n . When $n = 1$, this is true because a polynomial of degree $\leq q - 1$ is vanishing at q points. For the induction case, write $P = \sum_{j=0}^{q-1} P_j(x_1, \dots, x_{n-1})x_n^j$. For any fixed $x_1, \dots, x_{n-1}$, this vanishes for all x_n , so each P_j must vanish on every $x_1, \dots, x_{n-1}$. By induction, each P_j is 0, so P is 0. ■

Definition 4.9. A set $N \subseteq \mathbb{F}_q^n$ is *Nikodym* if, for all $x \in \mathbb{F}_q^n$, there is a line $L(x) \ni x$ with $L(x) \setminus \{x\} \subseteq N$.

Once again, because polynomials are restrictive on lines, it is smart to look at polynomials vanishing on N .

Theorem 4.10 (Dvir). *Any Nikodym $N \subseteq \mathbb{F}_q^n$ has size $\geq c_n q^n$ for a constant c_n depending only on n .*

Proof. Let N be Nikodym with $|N| < \frac{1}{(10n)^n} q^n$. There is a nonzero polynomial P vanishing on N of degree $\leq 2n|N|^{1/n} \leq q/5 < q - 1$. Let $x \in \mathbb{F}_q^n$, and let $L(x) \setminus \{x\} \subseteq N$. Then P vanishes at $\geq q - 1$ points on $L(x)$, and so vanishes on all of $L(x)$ since $\deg P < q - 1$. Thus, P vanishes on x . Since x was arbitrary, P vanishes on $\mathbb{F}_q^n$. But $\deg P < q - 1$, so $P = 0$, a contradiction. ■

Notice that we had a lot of slack in the above proof. We could have even only required that N contains half of $L(x)$ for each x , and the same proof would've worked. This shows just how restrictive polynomials are.

Definition 4.11. A set $K \subseteq \mathbb{F}_q^n$ is *Kakeya* if it contains a line in every direction. That is, for all $a \in \mathbb{F}_q^n \setminus \{0\}$, there is $b \in \mathbb{F}_q^n$ with $\{at + b : t \in \mathbb{F}_q\} \subseteq K$.

Theorem 4.12 (Dvir). *Any Kakeya set $K \subseteq \mathbb{F}_q^n$ has size $\geq c_n q^n$ for a constant c_n depending only on n .*

Proof. Let K be Kakeya with $|K| < \frac{1}{(10n)^n} q^n$. Then there is a nonzero P vanishing on K with $D = \deg P < q$. Write $P = P_D + Q$, where P_D is the degree- D component. Take $a \in \mathbb{F}_q^n \setminus \{0\}$ arbitrary, and let $b \in \mathbb{F}_q^n$ be such that $\{at + b : t \in \mathbb{F}_q\} \subseteq K$. Define $R(t) = P(at + b)$, so that R vanishes on $\mathbb{F}_q$ and $\deg R = \deg P < q$. Thus, $R = 0$. Now,

$$R(t) = P(at + b) = P_D(at + b) + Q(at + b),$$

so the coefficient of t^D in R is $P_D(a)$. Thus, $P_D(a) = 0$, but a was arbitrary, so P_D vanishes on $\mathbb{F}_q^n \setminus \{0\}$. Additionally, since P_D only contains terms of degree D , $P_D(0, \dots, 0) = 0$. But $\deg P_D < q$, so $P_D = 0$, a contradiction. ■

4.4 Roth's Theorem in $\mathbb{F}_3^n$

Definition 4.13. A set $C \subseteq \mathbb{F}_q^n$ is said to be a *capset* if $x + y + z \neq 0$ for any distinct $x, y, z \in C$.

In the case that $q = 3$, a set is capset iff it contains no 3-APs, $x, y, z \in \mathbb{F}_3^n$ are in AP iff $x + y + z = 0$. Because of this, we see that $\{0, 1\}^n$ is a capset, so we can find capsets of size 2^n . It has been shown that capsets of size 2.217^n exist. We can use polynomial methods to get an upper bound on the size of capsets.

Definition 4.14.

- We write $S_n \subseteq \mathbb{F}_q[x_1, \dots, x_n]$ for the subspace where each x_i has degree at most $q - 1$. Note $\dim S_n = q^n$.
- We write $S_n^d \subseteq S_n$ for the subspace where each poly has total degree at most d . We write $m_d = \dim S_n^d$.

Note that the evaluation map $S_n \rightarrow (\mathbb{F}_q)^{\mathbb{F}_q^n}$ is linear and that both spaces have dimension q^n . This is in fact an isomorphism, as $\prod_i (1 - (x_i - a_i)^{q-1})$ maps to the function $\begin{cases} 1 & x = a \\ 0 & x \neq a \end{cases}$, meaning evaluation is surjective, and so bijective.

Lemma 4.15. *Let $A \subseteq \mathbb{F}_q^n$ and $\alpha, \beta, \gamma \in \mathbb{F}_q$ with $\alpha + \beta + \gamma = 0$. If $P \in S_n^d$ satisfies $P(\alpha a + \beta b) = 0$ for all $a \neq b \in A^2$, then $|\{c \in A : P(-\gamma c) = 0\}|$ is at most $2m_{d/2}$.*

Proof. Let $M_n^d \subseteq S_n^d$ be the set of monomials of degree at most d (so that M_n^d forms a basis for S_n^d). We can split up the polynomial $P(\alpha x + \beta y)$ as

$$P(\alpha x + \beta y) = \sum_{\substack{m_1, m_2 \in M_n^d \\ \deg(m_1 m_2) \leq d}} c_{m_1, m_2} m_1(x) m_2(y).$$

Since $\deg(m_1 m_2) \leq d$ in each term, either $\deg(m_1) \leq d/2$ or $\deg(m_2) \leq d/2$ in each term. Thus, we can write

$$P(\alpha x + \beta y) = \sum_{m_1 \in M_n^{d/2}} m_1(x) F_{m_1}(y) + \sum_{m_2 \in M_n^{d/2}} m_2(y) G_{m_2}(x).$$

If we let B be the $A \times A$ matrix whose (a, b) entry is $P(\alpha a + \beta b)$, then B is diagonal, since $P(\alpha a + \beta b) = 0$ for $a \neq b$. Additionally, B is the sum of the $m_{d/2}$ matrices $(m_1(a) F_{m_1}(b))_{a, b}$ and the $m_{d/2}$ matrices $(m_2(b) G_{m_2}(a))_{a, b}$, each of which have rank 1 (since they are of the form uv^T for vectors u and v). Thus, B has rank at most $2m_{d/2}$. Since B is diagonal, this means there are at most $2m_{d/2}$ nonzero diagonal entries. But the diagonal entries are $P(\alpha c + \beta c) = \alpha(-\gamma c)$, so at most $2m_{d/2}$ of these are nonzero. ■

Theorem 4.16. *If $\alpha, \beta, \gamma \in \mathbb{F}_q$ are not all zero with $\alpha + \beta + \gamma = 0$ and $A \subseteq \mathbb{F}_q^n$ is such that $\alpha a + \beta b + \gamma c \neq 0$ for $(a, b, c) \in A^3 \setminus \{(a, a, a) : a \in A\}$, then $|A| \leq 3m_{(q-1)n/3}$.*

Proof. WLOG, $\gamma \neq 0$ (so in particular, multiplying by γ is a bijection). Let $0 \leq d \leq (q-1)n$, and write $V \subseteq S_n^d$ for the subspace of polynomials that vanish on $\mathbb{F}_q^n \setminus (-\gamma A)$. The subspace of S_n that vanishes on $-\gamma A$ has dimension $q^n - |A|$, so $\dim V \geq m_d - (q^n - |A|)$ (as we are taking the orthogonal complement within V). Let $P \in V$ have maximal support Σ . We must have that $|\Sigma| \geq \dim V$, as we could otherwise find some $Q \in V \setminus \{0\}$ vanishing on Σ , meaning that $P + Q$ has strictly larger support. Write $S(A) = \{\alpha a + \beta b : a \neq b \in A\}$, and note that $S(A) \cap (-\gamma A) = \emptyset$ by assumption. Thus, P vanishes on $S(A)$, so $P(-\gamma c) \neq 0$ for at most $2m_{d/2}$ elements $c \in A$. Thus, $|\Sigma| \leq 2m_{d/2}$. Thus,

$$m_d - q_n |A| \leq \dim V \leq |\Sigma| \leq 2m_{d/2},$$

meaning

$$|A| \leq (q^n - m_d) + 2m_{d/2}.$$

Now, $q^n - m_d$ is the number of monomials in S_n of degree $> d$. Note that there is a natural involution on monomials in S_n defined by $m \mapsto \frac{x_1^{q-1} x_2^{q-1} \dots x_n^{q-1}}{m}$ (i.e. replace each $x_{d_i}^i$ with $x_i^{q-1-d_i}$). This gives that $q^n - m_d$ is equal to the number of monomials in S_n of degree $< (q-1)n - d$, which is $\leq m_{(q-1)n-d}$. Thus, taking $d = 2(q-1)n/3$, $|A| \leq 3m_{(q-1)n/3}$. ■

Taking $q = 3$ and $\alpha = \beta = \gamma = 1$, this gives an upper bound on the size of capsets. In this case, we can compute that $|A| \in O(2.756^n)$, which is a result due to Ellenberg and Gijswijt.

4.5 Combinatorial Nullstellensatz

Theorem 4.17 (Combinatorial Nullstellensatz). *If $\mathbb{F}$ is a field, $d_1, \dots, d_n \geq 0$, $P \in \mathbb{F}[x_1, \dots, x_n]$ with $\deg P = \sum d_i$ and a nonzero $x_1^{d_1} x_2^{d_2} \dots x_n^{d_n}$ term, then P does not entirely vanish on any set of the form $E_1 \times E_2 \times \dots \times E_n \subseteq \mathbb{F}^n$ with $|E_i| > d_i$ for all i .*

Proof. We induct on $\deg P = \sum d_i$. When $\deg P = 0$, P is constant but has a nonzero $x_1^0 x_2^0 \dots x_n^0$ term, so P is constantly nonzero. Thus, it does not vanish on any subsets of $\mathbb{F}^n$.

Suppose $\sum d_i > 0$, and say WLOG $d_1 > 0$. Fix $E_1, \dots, E_n$ as in the theorem, and take $a \in E_1$. We can write $P = (x_1 - a)Q + R$ for $\deg Q = \deg P - 1$ and $R \in \mathbb{F}[x_2, \dots, x_n]$. If there is $a_2, \dots, a_n \in E_2 \times \dots \times E_n$ with $R(a_2, \dots, a_n) \neq 0$, then $P(a, a_2, \dots, a_n) = R(a_2, \dots, a_n) \neq 0$, and we are done. Otherwise, write $E'_1 = E_1 \setminus \{a\}$, and inductively find $(a_1, \dots, a_n) \in E'_1 \times E_2 \times \dots \times E_n$ with $Q(a_1, \dots, a_n) \neq 0$. We then have that $P(a_1, \dots, a_n) = (a_1 - a)Q(a_1, \dots, a_n) + R(a_2, \dots, a_n) = (a_1 - a)Q(a_1, \dots, a_n) \neq 0$. Thus, P doesn't vanish entirely on $E_1 \times \dots \times E_n$. ■

Theorem 4.18 (Cauchy-Davenport). *If p is prime and $A, B \subseteq \mathbb{F}_p$ are nonempty, then $|A+B| \geq \min(|A|+|B|-1, p)$.*

Proof. If $|A| + |B| > p$, then the sets A and $x - B$ must always intersect by the pigeonhole principle, meaning that $A + B = \mathbb{F}_p$ and we are done. Now, suppose that $|A| + |B| \leq p$, but that $|A+B| \leq |A| + |B| - 2$. We can take $C \supseteq A+B$ with $|C| = |A| + |B| - 2$. Now, define

$$P(x, y) = \prod_{c \in C} (x + y - c),$$

so that P vanishes on $A \times B$. Note that $\deg P = |A| + |B| - 2$. Additionally, the coefficient of $x^{|A|-1}y^{|B|-1}$ is $\binom{|A|+|B|-2}{|A|-1}$, which is not divisible by p , and so nonzero in $\mathbb{F}_p$. By the Combinatorial Nullstellensatz, P does not vanish entirely on $A \times B$, a contradiction. ■

We may think that there is slack here in that $\binom{|A|+|B|-1}{|A|-1}$ is also not divisible by p . However, the problem with this is that we need to have an $x^{|A|-1}y^{|B|-1}$ term to be able to get a contradiction, which cannot be improved upon.

4.6 Chevalley-Warning Theorem

If $\mathbb{F}$ is a finite field, then $\sum_{x \in \mathbb{F}} x^i = 0$ for $i < |\mathbb{F}| - 1$. This is because we can find some nonzero a with $a^i \neq 1$ (as $a \mapsto a^i - 1$ has at most $i \leq |\mathbb{F}| - 2$ roots), and

$$\sum_{x \in \mathbb{F}} x^i = \sum_{x \in \mathbb{F}} (ax)^i = a^i \sum_{x \in \mathbb{F}} x^i.$$

More generally, if $P \in \mathbb{F}[x_1, \dots, x_n]$ with $\deg P < n(|\mathbb{F}| - 1)$, then $\sum_{x \in \mathbb{F}^n} P(X) = 0$, as each monomial must have some x_i with degree $< |\mathbb{F}| - 1$ by the pigeonhole principle.

Theorem 4.19 (Chevalley-Warning). *If p is prime and $P_1, \dots, P_k \in \mathbb{F}_p[x_1, \dots, x_n]$ with $\sum \deg P_i < n$, then the number of common zeros of the P_i (i.e. $|\{x \in \mathbb{F}_p^n : P_1(x) = \dots = P_k(x) = 0\}|$) is divisible by p .*

Proof. Write

$$P(x) = \prod_{i=1}^k (1 - P_i(x)^{p-1}) = \begin{cases} 1 & P_i(x) = 0 \ \forall i \\ 0 & \text{else} \end{cases}.$$

Then $\deg P = (p-1)(\sum \deg P_i) < n(p-1)$, so $\sum_{x \in \mathbb{F}_p^n} P(x) = 0$. Thus, the number of common zeros of the P_i is 0 (mod p). ■

Theorem 4.20 (Erdos-Ginzburg-Ziv). *Let $n \in \mathbb{N}$ and $a_1, \dots, a_{2n-1} \in \mathbb{Z}$ (possibly with repeats). Then there is $I \subseteq [2n-1]$ of size n such that $\sum_{i \in I} a_i \equiv 0 \pmod{n}$.*

Equivalently, for any $a_1, \dots, a_{2n-1} \in \mathbb{Z}$, there is a subset of size n that has an integer average. Or, we could say that, for any $a_1, \dots, a_{2n-1} \in \mathbb{Z}/n\mathbb{Z}$, there is a subset of size n that sums to 0. Note also that the theorem is optimal: we can take $n-1$ 0's and $n-1$ 1's to find a counterexample of size $2n-2$.

Proof. We first prove this in the case $n = p$ is prime. Write $P_1(x_1, \dots, x_{2p-1}) = \sum a_i x_i^{p-1}$ and $P_2 = \sum x_i^{p-1}$. We have that $\deg P_1 + \deg P_2 = 2p-2 < 2p-1$, so the number of common zeros is divisible by p . Because 0 is a common root, there is at least one nonzero common root x . Since $P_2(x) = 0$, exactly p of the x_i are nonzero. Since $P_1(x) = 0$, this gives a subset of size p of the a_i that sums to 0.

Now, suppose that the theorem holds for $n = m$ and $n = k$, and let $a_1, \dots, a_{2mk-1} \in \mathbb{Z}$. We can construct disjoint $I_1, \dots, I_{2m-1} \subseteq [2mk-1]$ of size k such that $\sum_{i \in I_j} a_i \equiv 0 \pmod{k}$ (find a subset of size k using the theorem for $n = k$, discard those elements, and repeat until the theorem no longer applies. We can do this at least $2m-1$ times, as $(2mk-1) - (2m-2)k = 2k-1$). Write $S_j = \sum_{i \in I_j} a_i$. Then there is $J \subseteq [2m-1]$ of size m with $\sum_{j \in J} S_j \equiv 0 \pmod{m}$. But each S_j is divisible by k . Thus, if $I = \bigcup_{j \in J} I_j$, then $|I| = mk$ (as the I_j are disjoint), and

$$\sum_{i \in I} a_i = \sum_{j \in J} \sum_{i \in I_j} a_i = \sum_{j \in J} S_j \equiv 0 \pmod{km},$$

so the theorem holds for $n = mk$ as well. ■

4.7 Bounding Set Sizes

Definition 4.21. A set $X \subseteq \mathbb{R}^d$ is said to be one-distance if there is some $\alpha \in \mathbb{R}$ such that $|x - y| = \alpha$ for all distinct $x, y \in X$.

One-distance sets are not very interesting: if $X \subseteq \mathbb{R}^d$ is 1-distance, the maximum value of $|X|$ is $d+1$ (for example, take the vertices of a simplex). This becomes more interesting when we look beyond just the Euclidean distance.

Conjecture 4.22 (Petty). *If $\|\cdot\|$ is a norm on $\mathbb{R}^d$, then there exists a one-distance set of size $d+1$.*

This is not necessarily an upper bound in general: for example, the unit cube is one-distance under the supremum norm. Petty's conjecture is open for $d \geq 5$.

Definition 4.23. A set $X \subseteq \mathbb{R}^d$ is two-distance if there are $\alpha, \beta \in \mathbb{R}$ (not necessarily distinct) such that $|x - y| \in \{\alpha, \beta\}$ for distinct $x, y \in X$.

Every one-distance set is a two-distance set. A more interesting example is $\{e_i + e_j : i \neq j\} \subseteq \mathbb{R}_d$, where e_i denotes the standard basis vector.

Theorem 4.24. Every two-distance set in $\mathbb{R}^d$ has size $O(d^2)$.

Proof. Let $\{y_1, \dots, y_n\} \subseteq \mathbb{R}^d$ be two-distance with distances α and β . Suppose that $\alpha, \beta \neq 0$ (otherwise we have a one distance set, so that $n \leq d + 1$). Define polynomials

$$F_j(x) = (|x - y_j|^2 - \alpha^2)(|x - y_j|^2 - \beta^2) \in \mathbb{R}[x_1, \dots, x_n],$$

so that

$$F_j(y_k) = \begin{cases} \alpha^2 \beta^2 & k = j \\ 0 & k \neq j \end{cases}.$$

This in particular means that the F_j are linearly independent, as they have disjoint nonempty supports. If

$$P = \{1, x_i, x_i^2, x_i^3, x_i^4, x_i x_j, x_i x_j^2, x_i^2 x_j^2 : i, j \leq d\},$$

then each F_j lies in $\text{Span}(P)$, so $n \leq |P|$. Since $|P| = O(d^2)$, we are done. $\blacksquare$

Borsuk considered another question: if $X \subseteq \mathbb{R}^d$ is compact and has diameter 1, how many sets of smaller diameter are needed to cover it? In particular, he was interested in the ball $B = \{x \in \mathbb{R}^d : \|x\| \leq 1/2\}$. In this case, one can obtain a cover using $d + 1$ sets by drawing a regular simplex and saying $x \in A_i$ if x is closest to the vertex i . This is in fact optimal.

Theorem 4.25. Let $A_1, \dots, A_d \subseteq B$ be closed and cover B . Then there is some A_i with diameter 1.

Proof. Define $f : \partial B \rightarrow \mathbb{R}^{d-1}$ by

$$x \mapsto (\text{dist}(x, A_1), \dots, \text{dist}(x, A_{d-1})).$$

By Borsuk-Ulam, there is some x with $f(x) = f(-x)$. If there is some i with $d(x, A_i) = 0$, then $\text{dist}(-x, A_i) = 0$, and $x, -x \in A_i$, as A_i is closed. Otherwise, $x, -x \notin A_i$ for all $i \leq d - 1$, so $x, -x \in A_d$. $\blacksquare$

Borsuk conjectured that any $X \subseteq \mathbb{R}^d$ of diameter 1 can be covered with at most $d + 1$ sets of smaller diameter. However, Kahn and Kalai constructed a set where $1.2^{\sqrt{d}}$ sets are needed.

Theorem 4.26 (Frankl-Wilson). Let p be an odd prime, $n = 4p$, and $F \subseteq \{-1, 1\}^n$ so that no two vectors in F are orthogonal. Then,

$$|F| \leq 4 \left(\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{p-1} \right).$$

Proof. Let $G_1 \subseteq \{-1, 1\}^n$ consist of all vectors x that contain an even number of 1s and have $x_1 = 1$. Similarly, let G_2 contain all x with an even number of 1s and $x_1 = -1$, G_3 contain all x with an odd number of 1s and $x_1 = 1$, and G_4 contain all x with an odd number of 1s and $x_1 = -1$. We will show $|F \cap G_i| \leq \binom{n}{0} + \dots + \binom{n}{p-1}$.

We'll restrict to the case of G_1 , as they are all similar. Suppose that $y, y' \in G_1$ are distinct with $\langle y, y' \rangle \equiv 0 \pmod{p}$. Let

$$\begin{aligned} a &= \#\{i : y_i = 1, y'_i = 1\} \\ b &= \#\{i : y_i = 1, y'_i = -1\} \\ c &= \#\{i : y_i = -1, y'_i = 1\} \\ d &= \#\{i : y_i = -1, y'_i = -1\}, \end{aligned}$$

so that $\langle y, y' \rangle = a - b - c + d$. By assumption, $a + b$ and $a + c$ are even, so that $b + c$ is even. Thus, $2(b + c)$ is divisible by 4, and thus so is

$$n - 2(b + c) = (a + b + c + d) - 2(b + c) = a - b - c + d = \langle y, y' \rangle.$$

This means $\langle y, y' \rangle \in \{-4p, 0, 4p\}$. It is impossible for $\langle y, y' \rangle = -4p$ since they agree in the first coordinate, and impossible for $\langle y, y' \rangle = 4p$, since they are distinct. Thus, we must have $\langle y, y' \rangle = 0$.

Suppose now that $F \subseteq G_1$ has no two vectors orthogonal. For $y \in F$, define $P_y(x) = \langle x, y \rangle^{p-1} - 1$ over $\mathbb{F}_p$, so that $P_y(y') = 0$ for all $y' \neq y \in F$ and $P_y(y) = -1$. The collection $\{P_y : y \in F\}$ is therefore linearly independent in $\mathbb{F}_p[x_1, \dots, x_n]$. If we write the expression of P_y and replace each instance of x_i^{2k} with 1 and each instance of x_i^{2k+1} with x_i , we obtain a polynomial $\hat{P}_y$ that agrees with P_y on $\{-1, 1\}$ -vectors. Thus, the $\hat{P}_y$ are also linearly independent. Additionally, they are contained in the span of $1, x_i, x_{i_1}x_{i_2}, \dots, x_{i_1} \cdots x_{i_{p-1}}$, which has dimension $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{p-1}$, as desired. ■

The construction of Kahn and Kalai was to consider partitions $A \sqcup B = [n] = [4p]$ and let $S(A, B) = \{\{a, b\} \subseteq [n] : a \in A, b \in B\}$. We can think of indicator functions of these over $\binom{[n]}{2}$, which is equivalent to $\{-1, 1\}$ -vectors in $\mathbb{R}^{\binom{n}{2}}$. If $M = \{S(A, B) : |A| = |B|\}$, we may use Frankl-Wilson to bound the size of subsets of M that avoid having very little overlap (note $S(A, B) \cap S(C, D)$ is as small as possible when $|A \cap C| = p$), which is the same as avoiding being far apart. Considering the set of indicators of M in $\mathbb{R}^{\binom{n}{2}}$, one needs $1.2^{\sqrt{n}}$ sets of smaller diameter to form a partition.

4.8 Tutte-Grothendieck Invariants

One reason polynomials show up so much is due to so-called Tutte-Grothendieck invariants. These are properties that satisfy a deletion-contraction formula.

For example, let $\chi_G(k)$ denote the number of k -colorings of the graph G . This is a polynomial. The reason is that

$$\chi_G(k) = \chi_{G \setminus e}(k) - \chi_{G/e}(k),$$

where $G \setminus e$ is the graph with e deleted, and G/e is the graph with e contracted. Thus, noting that the number of k -colorings of a graph with 0 edges is $k^{\#V}$, one can induct on the number of edges to show that χ_G is a polynomial with degree $\#V(G)$.

Since χ_G is a polynomial, we can consider plugging in values other than nonnegative integers.

Definition 4.27. An acyclic orientation of a graph G is a choice of direction for each edge such that there are no directed cycles.

Given an acyclic orientation of $G \setminus e$, there is at least one acyclic orientation of G (if both possible orientations of e lead to cycles, they can be combined to find a cycle in $G \setminus e$). Both of the possible orientations work when contraction is possible without introducing cycles. Thus, if $A(G)$ is the number of acyclic orientations,

$$A(G) = A(G \setminus e) + A(G/e).$$

Noting additionally that there is 1 acyclic orientation of a graph with 0 edges, induction shows that the number of acyclic orientations of G is $(-1)^{\#V(G)} \chi_G(-1)$.

5 Ehrhart Theory

5.1 Pick's Theorem

Definition 5.1. A lattice polygon in $\mathbb{R}^2$ is a finite number of non-crossing segments meeting at every endpoint such that endpoints have integer coordinates.

Theorem 5.2 (Pick). If P is a lattice polygon, A_P is the area of P , b_P is the number of boundary points of P , and i_P is the number of interior points of P , then

$$A_P = i_P + \frac{b_P}{2} - 1.$$

Proof. Suppose first that Pick's Theorem holds for two polygons P and Q that only share boundary points. Then,

$$\begin{aligned}
A_{P \cup Q} &= A_P + A_Q = i_P + \frac{b_P}{2} - 1 + i_Q + \frac{b_Q}{2} - 1 \\
&= i_P + i_Q + \frac{b_{P \cup Q} + 2 \cdot \text{Shared Internal Boundary Points} + \text{Shared External Boundary Points}}{2} - 2 \\
&= i_P + i_Q + \frac{b_{P \cup Q} + 2 \cdot \text{Shared Internal Boundary Points} + 2}{2} - 2 \\
&= (i_P + i_Q + \text{Shared Internal Boundary Points}) + \frac{b_{P \cup Q}}{2} - 1 = i_{P \cup Q} + \frac{b_{P \cup Q}}{2} - 1.
\end{aligned}$$

Conversely, if it holds for $P \cup Q$ and P , then it holds for Q .

Note that it holds for rectangles with sides parallel to the axes: if the side lengths are a and b , then $A_R = ab$, $i_R = (a-1)(b-1)$, and $b_R = 2(a+b)$. Additionally, it must hold for right triangles with sides parallel to the axes: if it failed, it would have to fail by the same amount for the right triangle rotated by 180° , and would thus fail for their union, which is a rectangle with sides parallel to the axes. Now, for any triangle, one can draw a bounding rectangle and subdivide it into right triangles to see that Pick's Theorem holds for an arbitrary triangle.

Thus, to show that Pick's Theorem holds for all polygons, it suffices to show that one can always find a diagonal, as it allows for arbitrary lattice polygons to be triangulated. Let v be the leftmost vertex. If it is possible to draw the diagonal between the two neighbors, we are done. Otherwise, the leftmost point that is not v is not one of its neighbors (as there must be some edge that the diagonal intersects). Thus, connecting v to this point gives a valid diagonal. $\blacksquare$

5.2 The Ehrhard Polynomial

If $P \subseteq \mathbb{R}^2$ is a lattice polygon, the Ehrhart polynomial is

$$L : t \mapsto |(tP) \cap \mathbb{Z}^2|.$$

This is a polynomial, as

$$L(t) = i_{tP} + b_{tP} = A_{tP} - \frac{b_{tP}}{2} + 1 + b_{tP} = A_{tP} + \frac{b_{tP}}{2} + 1 = A_P t^2 + \frac{b_P}{2} t + 1.$$

We note in particular that $L(-1) = A_P - \frac{b_P}{2} + 1 = i_P$.

There is no generalization of Pick's Formula in higher dimensions. For example, if T_h is the convex hull of $(0,0,0)$, $(1,0,0)$, $(0,1,0)$, and $(1,1,h)$, then T_h will only ever contain those 4 lattice points. However, the area of T_h is $h/6$, which grows without bound. We can, however, salvage the Ehrhart polynomial.

Definition 5.3. A lattice polytope $P \subseteq \mathbb{R}^d$ is the convex hull of finitely many points in $\mathbb{Z}^d$.

The key Lemma of Ehrhart theory is that polynomials “do inclusion-exclusion for you”.

Lemma 5.4. If $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ and $d \in \mathbb{Z}_{\geq 0}$, the following are equivalent:

(a) For all $t \in \mathbb{Z}_{\geq 0}$,

$$\sum_{k=0}^{d+1} (-1)^{d+1-k} \binom{d+1}{k} f(t+k) = 0.$$

(b) f agrees with a polynomial of degree $\leq d$.

Proof.

(a) $\Rightarrow$ (b) We induct on d . When $d = 0$, the statement of (a) is that $f(t+1) - f(t) = 0$ for all t , meaning that f is constant,

and so agrees with a polynomial of degree 0. Now, suppose that (a) holds, and let $g(t) = f(t+1) - f(t)$. Then,

$$\begin{aligned}
0 &= \sum_{k=0}^{d+1} (-1)^{d+1-k} \binom{d+1}{k} f(t+k) = (-1)^{d-1} f(t) + \sum_{k=1}^d (-1)^{d+k-1} \binom{d+1}{k} f(t+k) + f(t+d+1) \\
&= (-1)^{d-1} f(t) + \sum_{k=1}^d (-1)^{d+1-k} \left(\binom{d}{k} + \binom{d}{k-1} \right) f(t+k) + f(t+d+1) \\
&= \sum_{k=1}^{d+1} (-1)^{d+1-k} \binom{d}{k-1} f(t+k) - \sum_{k=0}^d (-1)^{d-k} \binom{d}{k} f(t+k) \\
&= \sum_{k=0}^d (-1)^{d-k} \binom{d}{k} f(t+k+1) - \sum_{k=0}^d (-1)^{d-k} \binom{d}{k} f(t+k) \\
&= \sum_{k=0}^d (-1)^{d-k} \binom{d}{k} g(t+k).
\end{aligned}$$

By induction, g agrees with a polynomial of degree $\leq d-1$. Since $g(t) = f(t+1) - f(t)$, we may iterate to get $f(t+1) = \sum_{k=0}^t g(k) + f(0)$. Thus, it suffices to show that $\sum_{k=0}^t g(k)$ is a polynomial of degree $\leq d$. We may write g in the basis $\binom{k}{i}$ for $i \leq d-1$ and swap sums. Using the fact that $\sum_{k=0}^t \binom{k}{i} = \binom{t+1}{i+1}$, we see that f is indeed a polynomial of degree $\leq d$.

(b) $\Rightarrow$ (a) We induct on d . If $d = 0$, then f is constant, so that $f(t+1) - f(t) = 0$ for all t . Now, if f agrees with a polynomial of degree d , then $g(t) = f(t+1) - f(t)$ agrees with a polynomial of degree $d-1$. Thus, $0 = \sum_{k=0}^d (-1)^{d-k} \binom{d}{k} g(t+k)$. By the above computation, $0 = \sum_{k=0}^{d+1} (-1)^{d+1-k} \binom{d+1}{k} f(t+k)$ as well.

■

Theorem 5.5 (Ehrhart). *Let $P \subseteq \mathbb{R}^d$ be a full-dimensional lattice polytope. Then $L : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ by $t \mapsto |(tP) \cap \mathbb{Z}^d|$ is a polynomial of degree d , and the leading coefficient is the volume of P .*

Proof. It suffices to show this in the case that P is a lattice simplex, as subdividing into simplices and applying inclusion-exclusion yields the desired result for a general lattice polytope. By the lemma, we want to show

$$L(t+d+1) = \sum_{k=0}^d (-1)^{d-k} \binom{d+1}{k} L(t+k)$$

for all $t \geq 0$. Let $P = \text{conv}\{v_0, \dots, v_d\}$, write $Q_j = (t+d)P + v_j$, and let $Q = \bigcup_{j=0}^d Q_j$. We have that

$$\begin{aligned}
Q_j &= \left\{ v_j + \sum_{i=0}^d (t+d)\lambda_i v_i : \lambda_i \geq 0, \sum_{i=0}^d \lambda_i = 1 \right\} \\
&= \left\{ \sum_{i=0}^d a_i v_i : a_j \geq 1, a_i \geq 0, \sum_{i=0}^d a_i = t+d+1 \right\}.
\end{aligned}$$

If $\sum_{i=0}^d a_i v_i \in (t+d+1)P$, then $\sum_{i=0}^d a_i = t+d+1 \geq d+1$, so there is some j with $a_j \geq 1$. Thus, $(t+d+1)P = \bigcup_{j=0}^d Q_j = Q$. Additionally, for $J \subseteq \{0, \dots, d\}$,

$$\begin{aligned}
\bigcap_{j \in J} Q_j &= \left\{ \sum_{i=0}^d a_i v_i : a_i \geq \begin{cases} 1 & i \in J \\ 0 & i \notin J \end{cases}, \sum_{i=0}^d a_i = t+d+1 \right\} \\
&= (t+d+1 - |J|)P + \sum_{j \in J} v_j.
\end{aligned}$$

Thus, the number of lattice points in $\bigcap_{j \in J} Q_j$ is $L(t + d + 1 - |J|)$. By inclusion-exclusion,

$$\begin{aligned} L(t + d + 1) &= \#L(Q) = \sum_{k=1}^{d+1} (-1)^{k+1} \sum_{\substack{J \subseteq \{0, \dots, d\} \\ |J|=k}} \#L\left(\bigcap_{j \in J} Q_j\right) \\ &= \sum_{k=1}^{d+1} (-1)^{k+1} \binom{d+1}{k} L(t + d + 1 - k) \\ &= \sum_{k=0}^d (-1)^{d-k} \binom{d+1}{k} L(t + k), \end{aligned}$$

as desired. The fact that the leading coefficient is the volume of P comes from taking $t \rightarrow \infty$ in $L(t)/t^d$ and noting that it must be the volume. $\blacksquare$

What happens if we take $t = -1$? The equality $Q = (t + d + 1)P$ no longer holds. Rather,

$$dP \setminus Q = \left\{ \sum_{i=0}^d a_i v_i : 0 \leq a_i < 1, \sum_{i=0}^d a_i = d \right\}.$$

We can also think about the interior of P :

$$P^\circ = \left\{ \sum_{i=0}^d a_i v_i : 0 < a_i \leq 1, \sum_{i=0}^d a_i = 1 \right\}.$$

Thus,

$$-P^\circ + \sum_{i=0}^d v_i = \left\{ \sum_{i=0}^d a_i v_i : 0 \leq a_i < 1, \sum_{i=0}^d a_i = d \right\} = dP \setminus Q.$$

Now, the inclusion-exclusion formula for Q from above becomes

$$\#L(Q) = \sum_{k=1}^d (-1)^{d-k} \binom{d+1}{k} L(k-1),$$

as the $k = 0$ term corresponds to

$$\bigcap_{j=0}^d Q_j = \left\{ \sum_{i=0}^d a_i v_i : a_i \geq 1, \sum_{i=0}^d a_i = d \right\} = \emptyset.$$

Additionally, we can use the polynomial inclusion-exclusion on $L(d)$ to get

$$\begin{aligned} |-P^\circ \cap \mathbb{Z}^d| + |Q \cap \mathbb{Z}^d| &= |dP \cap \mathbb{Z}^d| = L(d) \\ &= \sum_{k=0}^d (-1)^{d-k} (-1)^{d-k} \binom{d+1}{k} L(k-1) = (-1)^d L(-1) + \sum_{k=1}^d (-1)^{d-k} \binom{d+1}{k} L(k-1). \end{aligned}$$

Thus,

$$(-1)^d L(-1) = |-P^\circ \cap \mathbb{Z}^d| = |P^\circ \cap \mathbb{Z}^d|.$$

More generally,

$$(-1)^d L(-t) = |(tP)^\circ \cap \mathbb{Z}^d|.$$

This is known as Ehrhart-MacDonald reciprocity.

5.3 Generating Functions

If $z = (z_1, \dots, z_d)$ and $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{Z}^d$, we'll write z^α for $z_1^{\alpha_1} \dots z_d^{\alpha_d}$. This allows us to associate a collection of lattice points with a formal series.

For $S \subseteq \mathbb{R}^d$, write

$$G_S(z) = \sum_{\alpha \in S \cap \mathbb{Z}^d} z^\alpha.$$

This need not converge, and so should not be treated like a function. However, we can restrict to a nice class of such series.

Definition 5.6. A Laurent polynomial is a sum

$$\sum_{\alpha \in S} c_\alpha z^\alpha,$$

where $S \subseteq \mathbb{Z}^d$ is finite and c_α are constants.

Definition 5.7. We say G_S is summable if there exists a Laurent polynomial g such that $g \cdot G_S$ is also a Laurent polynomial.

For instance, $G_{[0, \infty)}(z) = \sum_{k=0}^{\infty} z^k$ is summable using $g = 1 - z$.

If $w_1, \dots, w_n \in \mathbb{Z}^d$ are linearly independent, we'll let the simplicial cone be

$$K = \left\{ \sum_{i=1}^n \lambda_i w_i : 0 \leq \lambda_i \right\}$$

and the fundamental parallelepiped be

$$\Pi = \left\{ \sum_{i=1}^n \lambda_i w_i : 0 \leq \lambda_i \leq 1 \right\}.$$

Lemma 5.8. If $w_1, \dots, w_n \in \mathbb{R}^d$ are linearly independent, K is their simplicial cone, Π is their fundamental parallelepiped, and $v \in \mathbb{R}^d$, then

$$G_{v+K}(z) = \frac{G_{v+\Pi}(z)}{(1 - z^{w_1}) \dots (1 - z^{w_n})}.$$

Proof. For each $\alpha \in (v + K) \cap \mathbb{Z}^d$, we can write $\alpha = v + \sum_{i=1}^n \lambda_i w_i$. Writing $\lambda_i = \lfloor \lambda_i \rfloor + \{\lambda_i\}$, we have an expression

$$\alpha = v + \underbrace{\sum_{i=1}^n \{\lambda_i\} w_i}_{\in (v+\Pi) \cap \mathbb{Z}^d} + \underbrace{\sum_{i=1}^n \lfloor \lambda_i \rfloor w_i}_{\in \mathbb{Z}^d}.$$

Because the w_i are linearly independent, such a representation (i.e. a representation of the form $\alpha = p + \sum_{i=1}^n k_i w_i$ for $p \in (v + \Pi) \cap \mathbb{Z}^d$ and $k_i \in \mathbb{Z}$) is unique. Thus,

$$\frac{G_{v+\Pi}(z)}{(1 - z^{w_1}) \dots (1 - z^{w_n})} = \left(\sum_{p \in (v+\Pi) \cap \mathbb{Z}^d} z^p \right) \left(\sum_{k_1=0}^{\infty} z^{k_1 w_1} \right) \dots \left(\sum_{k_n=0}^{\infty} z^{k_n w_n} \right) = G_{v+K}(z).$$

■

If $P = \text{conv}\{v_1, \dots, v_n\} \subseteq \mathbb{R}^d$ is a polytope, then

$$G_P(\mathbb{1}) = |P \cap \mathbb{Z}^d|.$$

Writing $w_i = (v_i, 1) \in \mathbb{R}^{d+1}$, we have

$$\text{cone}(P) = \left\{ \sum_{i=1}^n \lambda_i w_i : \lambda_i \geq 0 \right\} \subseteq \mathbb{R}^{d+1}.$$

We can think of P as the intersection of $\text{cone}(P)$ with the hyperplane $\{x_{d+1} = 1\}$. More generally, tP is the intersection of $\text{cone}(P)$ with the hyperplane $\{x_{d+1} = t\}$. This means that terms in $G_{\text{cone}(P)}(z)$ with z_{d+1}^t correspond to lattice points in tP , so

$$G_{\text{cone}(P)}(z_1, \dots, z_{d+1}) = \sum_{t=0}^{\infty} G_{tP}(z_1, \dots, z_d) z_{d+1}^t.$$

In particular,

$$G_{\text{cone}(P)}(1, \dots, 1, z) = \sum_{t=0}^{\infty} L_P(t) z^t.$$

This is the Ehrhart series, denoted $E_P(z)$.

If we restrict to the case of a simplex (so that $n = d + 1$) with the v_i affinely independent, then

$$G_{\text{cone}(P)}(z) = \frac{G_{\Pi}(z)}{(1 - z^{w_1}) \dots (1 - z^{w_{d+1}})},$$

so that

$$G_{\text{cone}(P)}(1, \dots, 1, z) = \frac{G_{\Pi}(1, \dots, 1, z)}{(1 - z)^{d+1}}.$$

This gives that

$$E_P(z) = \frac{h_d^* z^d + \dots + h_1^* z + h_0^*}{(1 - z)^{d+1}},$$

where h_k^* is the number of lattice points in $\Pi \cap \{x_{d+1} = k\}$. Additionally,

$$\begin{aligned} E_P(z) &= \frac{h_d^* z^d + \dots + h_1^* z + h_0^*}{(1 - z)^{d+1}} = (h_d^* z^d + \dots + h_1^* z + h_0^*) \left(\sum_{t=0}^{\infty} \binom{t+d}{d} z^t \right) \\ &= h_d^* \sum_{t=0}^{\infty} \binom{t+d}{d} z^{t+d} + h_{d-1}^* \sum_{t=0}^{\infty} \binom{t+d}{d} z^{t+d-1} + \dots + h_1^* \sum_{t=0}^{\infty} \binom{t+d}{d} z^{t+1} + h_0^* \sum_{t=0}^{\infty} \binom{t+d}{d} z^t \\ &= h_d^* \sum_{t=d}^{\infty} \binom{t}{d} z^t + h_{d-1}^* \sum_{t=d-1}^{\infty} \binom{t+1}{d} z^t + \dots + h_1^* \sum_{t=1}^{\infty} \binom{t+d-1}{d} z^t + h_0^* \sum_{t=0}^{\infty} \binom{t+d}{d} z^t, \end{aligned}$$

so

$$L_P(t) = h_d^* \binom{t}{d} + h_{d-1}^* \binom{t+1}{d} + \dots + h_1^* \binom{t+d-1}{d} + h_0^* \binom{t+d}{d}.$$

More generally, if $\sum_{t=0}^{\infty} f(t) z^t = \frac{g(z)}{(1-z)^{d+1}}$ for a polynomial g with $\deg g \leq d$, then f is a polynomial. This provides an alternative proof of Ehrhart's Theorem.

Theorem 5.9 (Euler Relation). *Let P be a lattice polytope of dimension d , and let f_j be the number of faces of dimension j . Then $\sum_{j=0}^d (-1)^j f_j = 1$.*

Proof. We may count the number of lattice points in a polytope by counting the number of lattice points in the relative interior of each face. This gives

$$L_P(t) = \sum_{F \text{ face of } P} |(tF)^\circ \cap \mathbb{Z}^d| = \sum_{F \text{ face of } P} (-1)^{\dim F} L_F(-t).$$

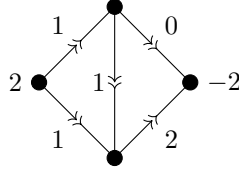
Taking $t = 0$,

$$1 = \sum_{F \text{ face of } P} (-1)^{\dim F} \cdot 1 = \sum_{j=0}^d (-1)^j f_j.$$

■

5.4 Applications of Ehrhart Theory

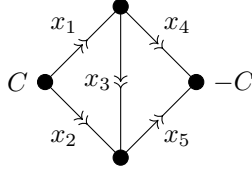
Suppose we have a directed graph G and we want to count integer flows:



That is, for some vertices s and t and some constant C , we want to count the number of functions $f : E(G) \rightarrow \mathbb{Z}_{\geq 0}$ such that

$$\sum_{e \ni s} f(e) = C \quad \sum_{e \text{ into } v} f(e) = \sum_{e \text{ out of } v} f(e) \quad \forall v \neq s, t.$$

In the above graph, we want to find x_i with



which translates into the equations

$$x_1 + x_2 = C \quad x_1 = x_3 + x_4 \quad x_2 + x_3 = x_5 \quad x_j \geq 0.$$

These are the same as

$$0 \leq x_2 = C - x_1 \quad 0 \leq x_4 = x_1 - x_3 \quad 0 \leq x_5 = C - x_1 - x_3,$$

which determine a lattice polygon in the plane. Thus, we want the number of lattice points in a dilate of a particular polygon, meaning that the number of integer flows can be counted by a polynomial in C . This same argument holds in general.

If G is a graph on n vertices, let

$$P_G = [0, 1]^n \setminus \bigcup_{(i,j) \in E(G)} \{x : x_i = x_j\}.$$

Then, the number of lattice points in cP_G is the number of $(c+1)$ -colorings of G , so the number of colorings of G can also be counted by a polynomial.

If

$$\Delta_d = \{x \in \mathbb{R}^d : 0 \leq x_1 \leq x_2 \leq \cdots \leq x_d \leq 1\},$$

then integer points in $n\Delta_d$ correspond to choices of $0 \leq x_1 \leq \cdots \leq x_d \leq n$, which correspond to multisets of $[n+1]$ of size d . Integer points in $(n\Delta_d)^\circ$ correspond to choices of $0 < x_1 < \cdots < x_d < n$, which correspond to subsets of $[n-1]$ of size d . In particular, $L_{\Delta_d}(-n) = \binom{n-1}{d}$.

If $G = ([n], E)$ is a simple graph, $e_1, \dots, e_n \in \mathbb{R}^n$ are the standard basis vectors, and $[e_i, e_j]$ denotes the segment connecting e_i and e_j , the zonotope is

$$Z_G = \sum_{(i,j) \in E} [e_i, e_j],$$

where the sum denotes a Minkowski sum. We have $L_{Z_G}(t) = c_{n-1}t^{n-1} + \cdots + c_1t + c_0$, where c_k is the number of forests in G on k edges. Z_G is $n-1$ dimensional because it is normal to the vector $(1, \dots, 1)$. In the case $G = K_n$, Z_G is called a permatohedron, and is the convex hull of $(\pi(1) - 1, \dots, \pi(n) - 1)$ for all $\pi \in S_n$.

5.5 Geometry of Numbers

We say $L = A[\mathbb{Z}^n]$ is a rank- n lattice if A is an invertible $n \times n$ matrix. We write $d(L) = |\det A|$.

Theorem 5.10 (Minkowski). *If C is convex, bounded, and centrally symmetric in $\mathbb{R}^n$, and L is a rank- n lattice with $\text{vol}(C) > 2^n d(L)$, then C contains a nonzero point of L . If C is compact, this is true when $\text{vol}(C) = 2^n d(L)$.*

Proof. Let $v_1, \dots, v_n$ be a basis of L and $F = \{\sum_{i=1}^n \lambda_i v_i : 0 \leq \lambda_i < 1\}$, so that $\text{vol}(F) = d(L)$ and $\mathbb{R}^n$ is partitioned by translates of F . If $\text{vol}(C) > 2^n d(L)$, then $\text{vol}(\frac{1}{2}C) > d(L)$. For $v \in L$, let $S_v = (\frac{1}{2}C) \cap (v + F)$. These are pairwise disjoint, and the nonempty ones partition $\frac{1}{2}C$, so $\sum_{v \in L} \text{vol}(S_v) = \text{vol}(\frac{1}{2}C) > d(L)$. If $S'_v = S_v - v \subseteq F$, then $\text{vol}(S'_v) = \text{vol}(S_v)$, so $\sum_{v \in L} \text{vol}(S'_v) > d(L) = \text{vol}(F)$. Thus, there are $u \neq v \in L$ such that $S'_v \cap S'_u \neq \emptyset$, say $a \in S'_v \cap S'_u$. Then $a + v, a + u \in \frac{1}{2}C$. Since C is symmetric, $-a - u \in \frac{1}{2}C$, so convexity means $\frac{v-u}{2} \in \frac{1}{2}C$, so $v - u \in C$. But $v \neq u$, so $v - u \in L \setminus \{0\}$, as desired. ■

Theorem 5.11. *If $p = 4k + 1$ is prime, then p can be written as the sum of two squares.*

Proof. Since $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic, there is $u \in \mathbb{Z}$ with $u^2 \equiv -1 \pmod{p}$. Let $M = \begin{pmatrix} 1 & 0 \\ u & p \end{pmatrix}$ and $L = M[\mathbb{Z}^2]$, so that $d(L) = \det M = p$. If $(x, y) = M(t_1, t_2)$, then $x^2 + y^2 = t_1^2 + (ut_1 + pt_2)^2 \equiv 0 \pmod{p}$. If $B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 2p\}$, then $\text{vol}(B) = \pi(\sqrt{2p})^2 > 4p$. Thus, B contains a nonzero point (x, y) of L . Now, $x^2 + y^2 < 2p$ is nonzero and divisible by p , so $x^2 + y^2 = p$. ■

Theorem 5.12. *Every positive integer can be written as the sum of 4 squares.*

Proof. It suffices to show this for primes. If $p > 2$ and $a \in \mathbb{Z}$, there are $r, s \in \mathbb{Z}$ such that $r^2 + s^2 \equiv a \pmod{p}$ (there are $\frac{p+1}{2}$ squares and the map $t \mapsto a - t$ is injective, so the functions r^2 and $a - s^2$ both take on $\frac{p+1}{2}$ values, meaning they must coincide at least once). Take $r^2 + s^2 \equiv -1 \pmod{p}$, and let

$$M = \begin{pmatrix} p & 0 & r & s \\ 0 & p & s & -r \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad L = M[\mathbb{Z}^4].$$

Then $d(L) = \det M = p^2$. If $(x_1, x_2, x_3, x_4) = M(t_1, t_2, t_3, t_4)$, then

$$\begin{aligned} x_1^2 + x_2^2 + x_3^2 + x_4^2 &= (pt_1 + rt_3 + st_4)^2 + (pt_2 + st_3 - rt_4)^2 + t_3^2 + t_4^2 \\ &\equiv (r^2 + s^2 + 1)t_3^2 + (r^2 + s^2 + 1)t_4^2 + (2rs - 2rs)t_3 + t_4 \equiv 0 \pmod{p}. \end{aligned}$$

Let $B = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1^2 + x_2^2 + x_3^2 + x_4^2 < 2p\}$, so that $\text{vol}(B) = \frac{\pi^2}{2}(\sqrt{2p})^4 = 2\pi^2 p^2 > 16p^2$. Then B contains a point of $L \setminus \{0\}$, meaning that p can be written as the sum of 4 squares. ■